Manual

Data Collection/Information Functions for ERP Batches, MES Batches, Serial Numbers

AIP-TRT 8.1

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Contents

1 Overview of Data Collection/Information Functions for ERP Batches, MES Batches, Serial Numbers

Purpose

The AIP features contained in this function package make it possible to enter batch-related data directly in production using shop floor terminals or data acquisition PCs.

Integration

Data entered via the AIP terminal can be displayed and/or evaluated in different MOC applications. The collected data can be uploaded via relevant interfaces.

Functions

Order-related data acquisition and posting functions

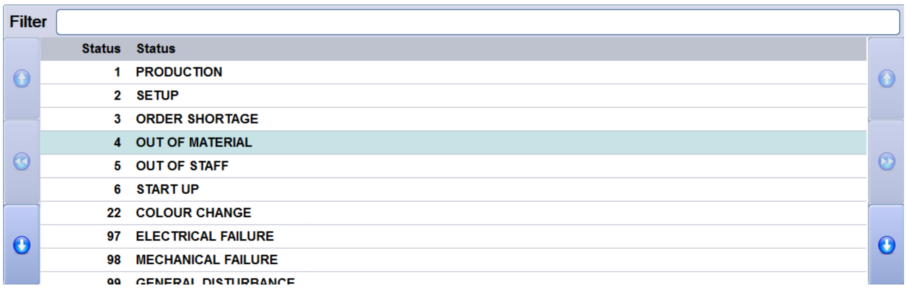
- Input batches can be entered at the same time as operations are logged on (configurable per operation and workplace)
- Produced output batches can be entered at the same time as operations are logged on
- Changing of input and output batches can be recorded and checked for validity while the operation is running and/or the order is being processed
- Entry of goods receipt batches
- Automatic generation of material movements (incoming goods/outgoing goods) depending on the receipt or consumption of input and/or output batches
- Batch-related collection of quantities and time for produced batches of material
- Entry of ERP or MES batch numbers using the keyboard and/ or barcode
- A validation check to ascertain if documentation is required (documentation obligation) for consumable material when the operation and/or the input batch is logged on
- A ticket/ label is automatically printed in the HYDRA standard format using the assigned printer when a new ERP or MES batch is generated

2 Operation of AIP

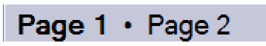
2.1 Special control and display elements within AIP

Tables

Tables are displayed in a uniform way within AIP. This affects the basic display (workplaces, operations, …) as well as the selection lists of posting dialogs.



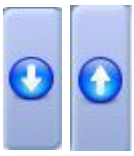
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE



Provided that information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. By clicking/touching the user can directly switch to another page.

An operation may be selected using the mouse, touch screen, keyboard (arrow keys:'↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Please find the following example: When an operation is logged on, those operations may be selected that are included in the sequencing list or that are planned for the corresponding workplace or group. However, when operations are interrupted, only running operations may be selected.

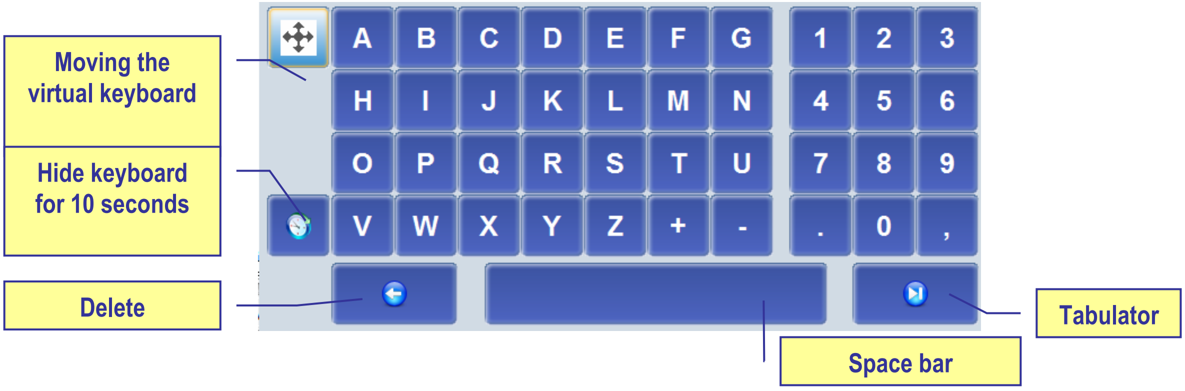

Scrolling page by page (up or down) in the table.


Scrolling to the left or right. Only those buttons are activated that are reasonable for the current situation. This figure shows that scrolling to the left has been deactivated.

A “table filter” may optionally be displayed (customizing). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is case-insensitive.

Virtual keyboard

The virtual keyboard allows for data to be entered manually via touch screen or a connected mouse. To make it easier for inexperienced users to find the required keys, the numeric key pad is organized like the telephone and letters are aligned in alphabetical order. Consequently, both differ from the computer keyboard which usually is aligned in the “QWERTZ keyboard layout”. The virtual keyboard is displayed automatically as soon as an input field is focused.



The driver needs to be configured respectively for the touch screen to be able to move the keyboard (settings in the control panel of the terminal/PC)! The virtual keyboard only supports the characters "0" - "9", "A" - "Z" and "+“, "-“ , ".“ and ",“. Other characters or languages are not supported. It is recommendable to use an additional keyboard if texts in other languages have to be entered.

The start position of the virtual keyboard can be defined by a setting in the configuration file keyboard.ini. Subject to the screen resolution, the parameters xpos= and ypos= need to be enabled in the configuration file.

If the virtual keyboard is not to be shown in general, the parameter –t needs to be included in the parameter bar parameters= of the configuration file ctaip.ini.

Date display

AIP supports a country-specific date format in dynamic dialogs. This can be configured in the "control panel", "regional settings", "short date" dialog of the terminal/PC. The following has to be taken into account in this context:

- Years are always four characters long.
- Months and days are always 2 characters long.
- “-“, “/“ and “.“ are allowed separators
- Blanks must not be included in the “short date” format, i.e. the <BLANK> separator is not allowed.
- The date separator “.” (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

English(USA)	MM/DD/YYYY
Danish	MM-DD-YYYY
Customer-specific 1	YYYY-MM-DD
Customer-specific 2	YYYY/MM/DD
Customer-specific 3	MM/YYYY/DD

Please note

If the date format does not correspond to conventions a note appears when the program is started and the date format is set to MM/DD/YYYY.

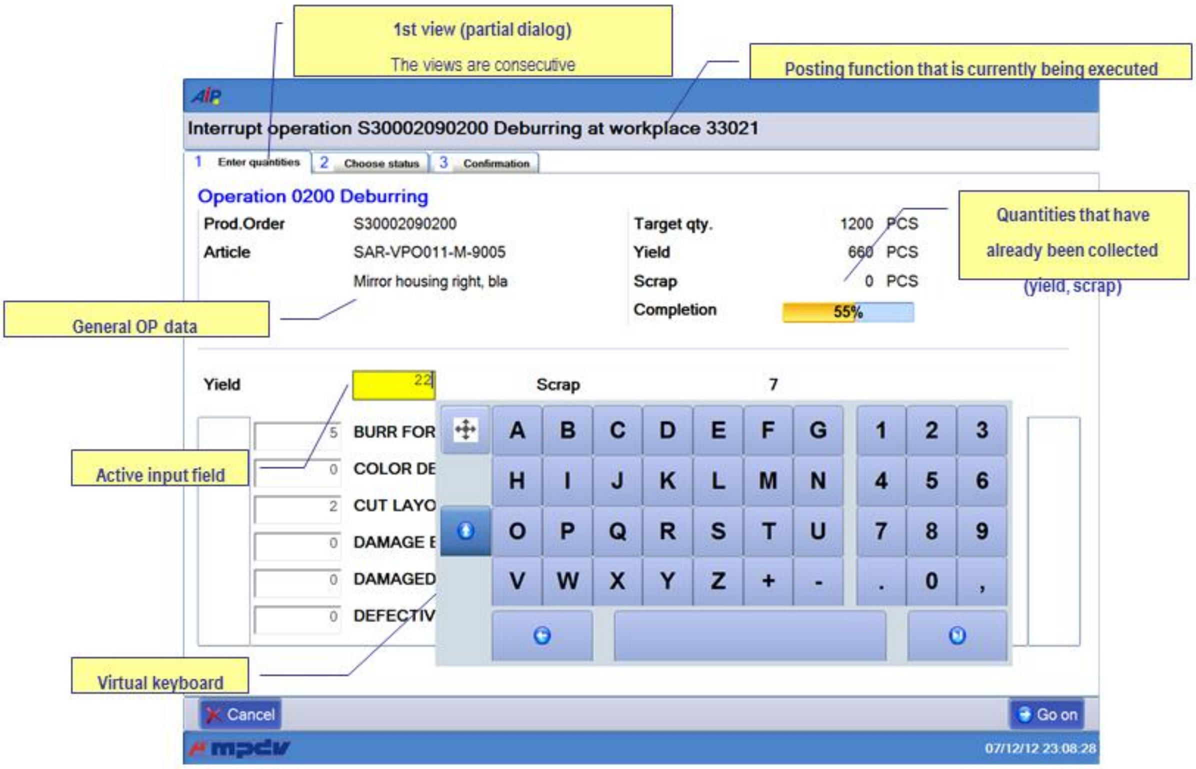
The year is displayed only by two characters in the status bar.

2.2 General description of the posting process with AIP

In general, posting dialogs are divided into several visual views at AIP. These views (partial dialogs) cover the entire screen and only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. This process is described by way of the following example (interrupt operation). The other dialogs can be operated in the same way.

The “interrupt operation” function is executed. This task is started by clicking the “interrupt operation” function from the second toolbar:

The “interrupt operation” dialog opens and the first view is displayed. The function that is currently being executed (in this case: interrupt operation) is shown in the header.



The first view “enter quantities” provides the user with the possibility to enter the produced yield or scrap quantities. The virtual keyboard is shown or hidden automatically, subject to the active input field.

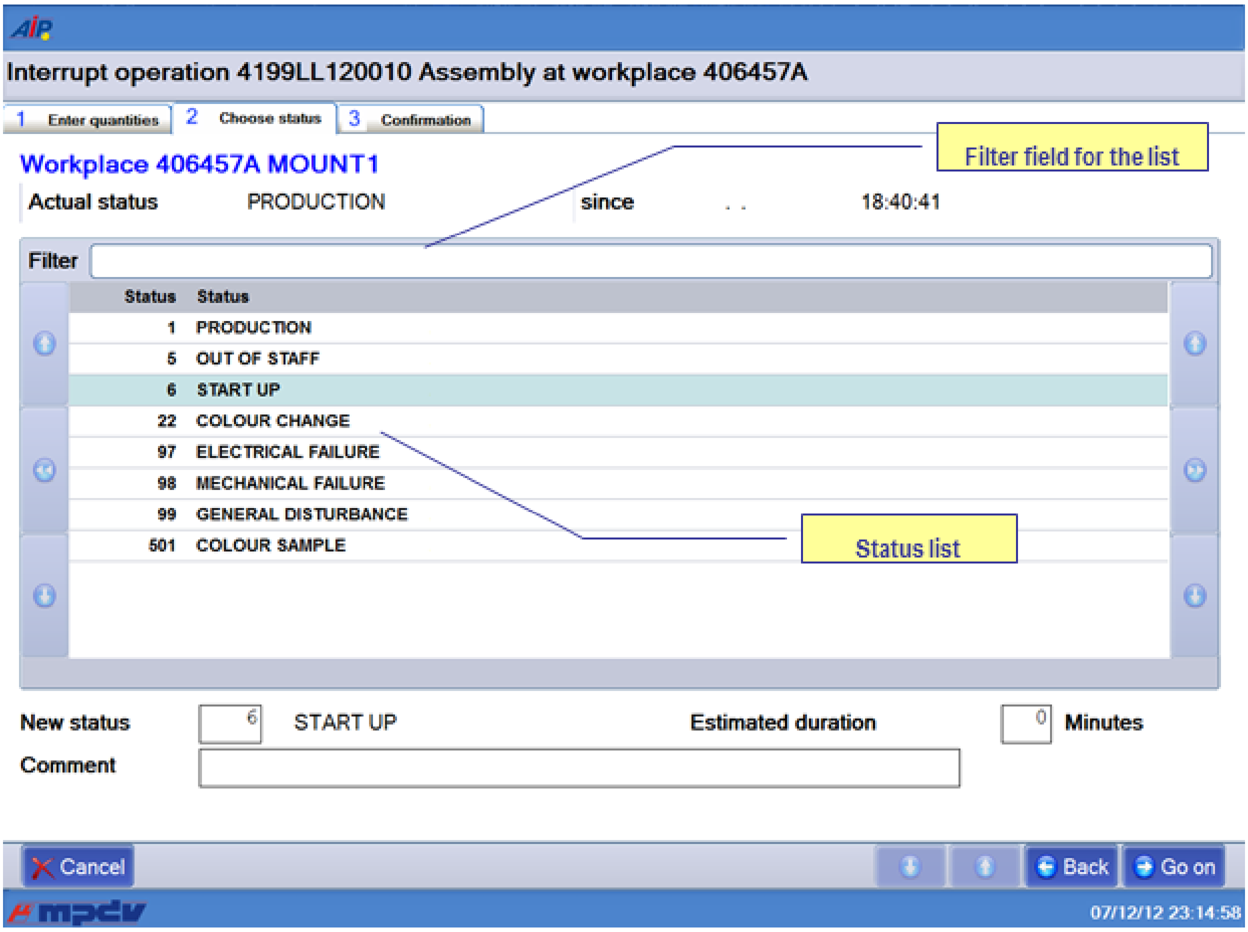
Quantities can be entered using the virtual keyboard or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). Once all values have been entered in the first view, the next view can be opened by clicking the “next” button.

The “cancel” button is displayed in all partial dialogs and allows for the entire posting dialog to be cancelled/closed at any time.

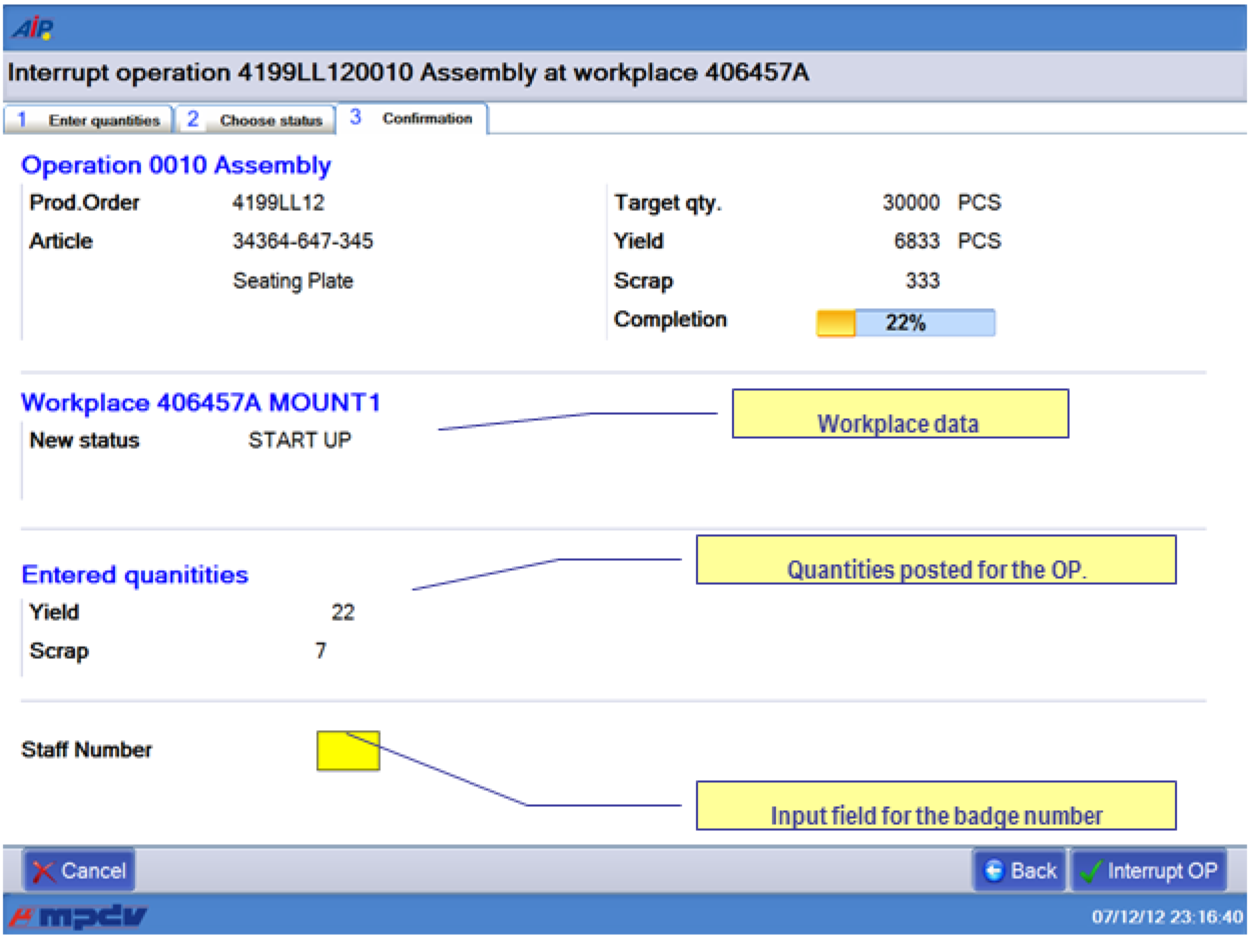
The next view can be opened either by clicking the “next” button or by clicking another tab (in our example: “select status” or “confirm”). Please note in this context, that no view can be skipped when the views are navigated bottom up (view 1 → view 2 → view 3). This means: if you are in the first view (enter quantities) and you click the third view (confirm), the second view (select status) will be displayed first.

Vice versa, when navigating top down (e.g. from the “confirm” view to the “enter quantities” view), every view may directly be opened by clicking at it. In this case, views are actually skipped. But the “back” button also allows for the views to be opened one after the other (top down).

As long as the dialog has not been confirmed, entered data may be changed at any time by scrolling back and forth.



The workplace status that is to be set, once the operation has been interrupted, is determined in the second view “select status”. This status may be chosen from the displayed status list. This list can be restricted using the “filter” field. Once the required values have been entered, the next view can be opened by clicking “next” (in our example it is the last view).



The partial dialog “confirm” shows a summary of all values entered so far in the dialog. Provided that the user agrees with the entered data, the “interrupt operation” dialog can be confirmed, once the badge number has been entered. Then the dialog including the data is sent to the server and posted.

If an input field of the dialog is not filled out properly (e.g. a mandatory field is empty) the field is highlighted in red in the corresponding view and focused to enable the user to directly correct the field content.

If a workflow dialog is opened it may directly be exited by clicking the ESC button. This is also the case, if the virtual keyboard is opened. Thus, the ESC button cannot be used to close the virtual keyboard.

3 Basic AIP Display

	→	

In general, *AIP* has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons within the touch screen or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these selection lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.

To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

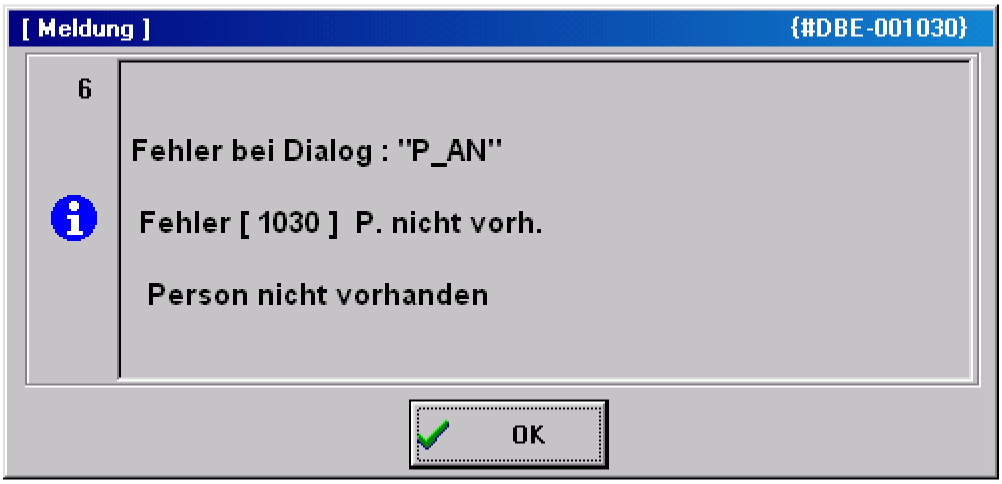
3.1 Basic displays – header and footer

Header

The *AIP* logo is displayed top left of the screen, which may be exchanged by a customer logo after corresponding configuration.

Possible messages (e.g. if a dialog is opened for more than five minutes) are displayed to the right of it.

A separate window opens to display error messages that occur during data collection (e.g. validity checks).



Basic displays

A maximum of 16 workplaces or machines can be assigned to the *AIP* terminal. The individual workplaces can be found within the list area in the order in which they have been assigned to the terminal in MOC.

As regards the basic display of the *AIP* terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured within the terminal configuration at the client. The individual basic views are described in the sections that follow.

Footer

The MPDV logo can be found at the bottom left of the screen. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

AIP status

	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.

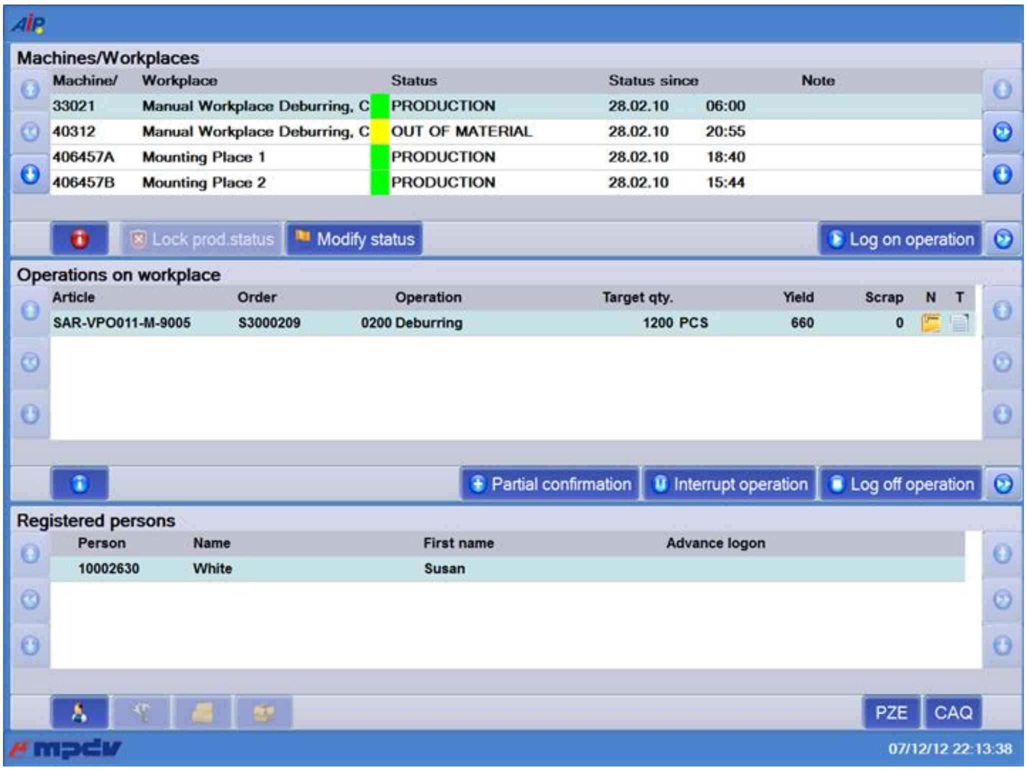
	Commands are being transferred to the server
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

3.2 Basic display “tabular view“

1st list Workplaces assigned to the terminal

2nd list List of registered operations

3rd list (optional) e.g. list of registered staff



The tabular basic display consists of two or three tables, subject to the configurations made. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional display).

“Machines/workplaces" table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as the designation are displayed.

Status

The status is displayed in color and as status text. Coloring is as follows:

- green: Production
- yellow: Assigned status
- red: Not assigned

Status since

Point in time since the status is available.

For ADE workplaces[1] the point in time refers to the last manual status change, for MDE workplaces it refers to the time when the last status change was identified (for machine connections), to the point in time when the status was changed manually the last time or to the time of the last shift change.

Please note

It is indicated here if the “block production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.

Please note

By way of “customizing” services it is possible to adapt the layout of display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic display of terminals. The software does not allow it.

Provided that the "compensate manual quantities" option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately, once they have been entered but only once lists have been reloaded.

[1] We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an ADE workplace.

"Operations at workplace" table

The second table shows the operations that are currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number.*

Target quantity

Target quantity that is defined for the operation.

Yield

Yield which has already been produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity which has already been produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note, which should be visible at the terminal, has been recorded for this operation in the graphic planning board at the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a line that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured respectively. Which information is displayed in this list depends, among other things, on the workplace configuration.

The following lists can be displayed:

- Staff logged on to the currently selected workplace (BDE)
- Resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow for switching between these lists.

Please note

The registered staff displayed in the third list correspond to the list of the dialog “F5 registered persons…”. Selecting a person in the third list does *not* affect selection of the operation in the list of OPs running at the workplace and, as a result, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic display

A toolbar, which may be configured by customizing, is assigned to each list in the basic display. This makes the purpose of the function clear to the user. The “partial upload/confirmation” function can be found, for example, below the list of registered operations.

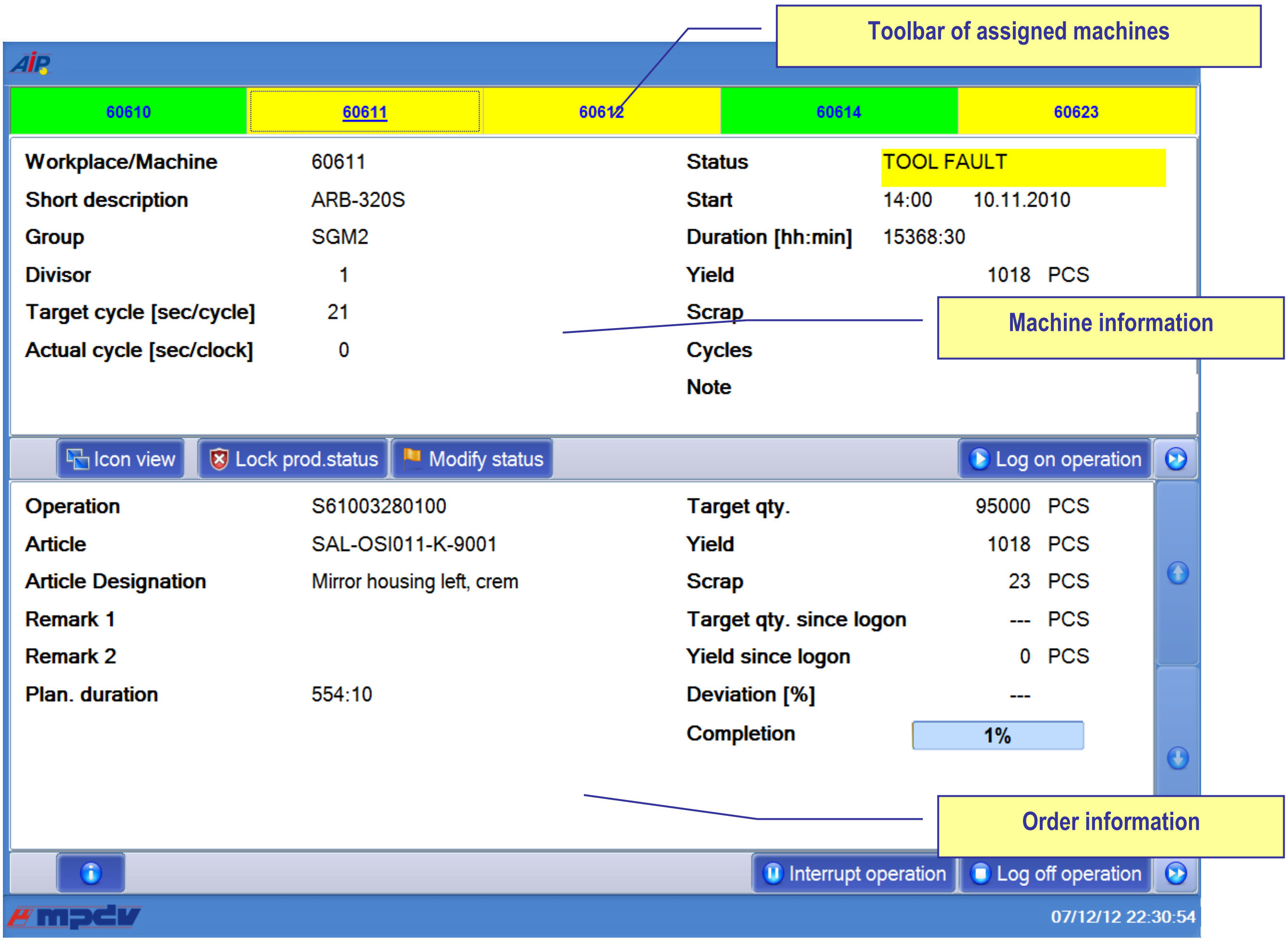
In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. change partitioning) can be opened by clicking the corresponding button.

Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

3.3 "Machine overview" basic display

If the “change view” button is clicked in the basic display, the view changes to the following presentation:



This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation altogether has three areas:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines as well as to shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. Provided that several orders/OPs are logged on, it can be switched between them using arrow keys that are then shown additionally.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$Displayed\ partitioning = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \ldots \right]$$

TLG_M	Partitioning of the machine (TLG in mnr.lst)
DIV_M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG_{AGi}	Partitioning of the individual order (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview at the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal may get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) at the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time in which the production lock of the machine has not been active). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:
TargetSinceLogon = NetRunningTime[sec] * Partitioning/TargetCycle[sec/stroke]
NetRunningTime: Time since logon, in which the production lock has not been set. This calculation does not take into account any breaks that might be defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

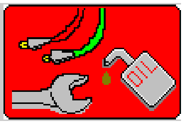
Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced “since logon”.
Calculation: Deviation[%] = 100% * (YieldSinceLogon - TargetSinceLogon) / TargetSinceLogon

Completion

The bar represents the proportion of “yield”, which has been produced until now, compared to the “target quantity”.

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced by a picture showing a yellow or red oilcan, depending on the maintenance activity that is required:



3.4 “Machines as icons” basic display

This view can be configured as the default display within the terminal configuration at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green:	Production
- yellow:	Assigned status
- red:	Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon presentation of machines”.

4 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
--------	--------------------	**---- General ---**
00.	00.ABC123	Data not defined, ➔ 00. will be deleted and data “ABC123” will be passed to standard processing
01.	01.OK 01.ESC	Action barcode ➔ Dialog cancelled or ended with OK button or Esc button.
--------	--------------------	**---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---**
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Suborder number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16.200	Machine → Acronym <MNR> ➔ Passed to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> ➔ Passed to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> ➔ Passed to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > ➔ Passed to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> ➔ Passed to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
--------	--------------------	**----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---**
40.	40.100 40.MONTAGE	Destination → Acronym <ZLO> ➔ Passed to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> ➔ Passed to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→➔
43.		Throughput batch number → Acronym <DLL>→➔
44.		Alternative batch number → Acronym <CNR:ALT1>→➔
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
--------	--------------------	**---- Mainly for the HYDRA-PZE module ---**

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> ➔ Passed to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> ➔ Passed to dialog with FGR=1 or FGR=1001
--------	**-------------------**	**---- Customer-specific barcodes ---**
90.		
…		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]
BarcodePrefixChar=$

If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font “Codedreineun” and prefix “.“

Prefix	Barcode	Raw data
10.	*10.123456780100* *10._ABCD12340100*	ANR = 123456780100 ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

4.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR **BarKenn91=EGR:GUT**	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345“ → SAPCNR=12345

Firmly assigned barcode prefixes:
10:ANR
11:AUNR
12:AFOLG
13:AGNR
14:UAGNR
15:RMNR
16:MNR

17:MST
18:AUSGRD
19:AGGGRD
20:BPOS
21:LPKZ
22:SPLNR
40:ZLO
41:TPE
42:CNR
43:DLL
44:CNR:ALT1
45:CNR:ALT2
46:CNR:ALT3
47:CNR:ALT4
48:CNR:ALT5
49:CNR:ALT6
50:KNR
51:PNR
52:KST
53:FGR

5 Local Configuration File ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the c:\ctaip directory.

Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

5.1 Basic configuration

Entry	Comment
Section [system]	

Entry	Comment
Section [system]	
`Usr=21`	Distinct terminal number

`Hypath=d:\hydra\`
 	
`Hypath=/usr/hydra/`
	HYDRA server path:
	
	Windows NT:
	-> DOS notation, the drive is the local drive of the server on which HYDRA or xMES is installed
	Unix:
	->Unix notation

`Hostname=192.9.200.24`	Internet address of the server
`Offlinetimeout=600`	In offline mode, the interval after which online access should be attempted the next time. The interval is specified in seconds
`Showcursor=on`	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
`Loadfile=` `ctnet\win\ctaip.txt`	Configuration file for downloading the application from the server. The path is relative to the server directory (i.e. within “hypath”)
`Watchdog=on`	ON: Watchdog is activated OFF: Watchdog is not activated
`Demo=off`	‘on’: Offline demo mode; always off in the production environment
`parameters=-t`	The –t parameter switches off the virtual keyboard
`TMOUT_C=xxx`	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_S=xxx`	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_R=xxx`	Timeout for RECEIVE of the server If not specified, default = 120 seconds

Entry	Comment
`TMOUT_F=xxx`	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	**Configuration of customized barcode prefixes.**

`BarKenn90=MNR4`
`...`
`BarKenn99=ANR3`
	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.

Section [comports]	

`Com1=0`
`com2=MSS`
`com3=BAR`
`Com3=LEGIC`
`Com4=RFLESER`
	Assignment of serial interfaces to the connected devices
	MSS – machine interface
	BAR, LEGIC, RFLESER – various reading devices

Section [MSS-INIT]	
`ZAEHLER=|1|2|3|4|5|6|7|8|`	Assignment of physical inputs of the MSS (machine interface) to logical counters (ZAEHLER) according to configuration: The first connector (labeled “0” on the MSS) corresponds to the logical counter no. 1 Please note: MSS1 has only 8 inputs If digital inputs are also to be used with MSS1, configuration should be changed as follows: ZAEHLER=|1|2|3|4| IN=|5|6|7|8|
`IN=|9|10|11|12|13|14|15|`	Assignment of physical MSS inputs to logical inputs as per configuration: The ninth connector (labeled “8” on the MSS) corresponds to the logical input no. 1
`CHARGE=|5|6|`	For batch recording: Inputs for automatic batch changes In this case, the connectors 5 and 6 are the inputs for the automatic batch change.
`MSSZyklusBerechnung=ON` `MSSZyklusReferenz=0`	Activates reading out of cycle time values from the machine interface (MSS). Reference for cycle calculation (smallest time unit of the MSS). The default value is 0, if the parameter is not specified. 0 corresponds to 100 ms. 2 corresponds to 20 ms.
`CalculateCycle=ON`	The terminal itself calculates the actual cycle. If the connected control does not provide a determined actual cycle, then a calculated actual cycle can be displayed: This calculated value is also available for DS100 terminals. Please note: The calculation cannot provide exact values.
`WochenEnde_ProdCheck=ON`	This function prevents the weekend automatic from affecting the “production“ status and the workplace from being set to status 999. ON is set by default In case WochenEnde_ProdCheck=OFF, the automatism switches to status 999.

Entry	Comment
`sFrom999ToNotAttributed=OFF`	Only affects HYDRA-MDE machines with operation mode = "no monitoring". If the weekend automatic function is enabled this option prevents the machine from switching to the "not assigned" status when the weekend automatic ends. Reasons: The "not assigned" status may not be set manually for machines that are configured with the "no monitoring" option. The "not assigned" status is normally only set for machines with operation mode = "cyclic monitoring" or "monitoring by operating signal".
Section [ext. software]	

`Button=Editor`
`WindowName=Editor`
`SearchParts=On`
	Configuration of the button in the top line: A previously started program can be called to the foreground at the push of a button.
	
	Button: button caption
	WindowName: Name of the program (e.g. from the taskbar).
	SearchParts=On: parts of WindowName are sufficient
	SearchParts=Off: WindowName must be entered completely.
	The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.

`ProgFileName=c:\Programme\wincmd\Wincmd32.exe`	The program that is started if the program mentioned above cannot be called to the foreground.
`AutoStart=on`	This option starts the program (ProgFileName) when starting the terminal program.
Section [PDV]	
`Modus=PDV` `Modus=PDV,BDE`	Operation as interface IOP←→DOS terminal (standard) Operation as shop floor terminal with PDV
`Terminals=121,122,123`	CTDOS terminals “connected” by LAN. Only required with Mode=PDV
`PDVTerminalDir=c:\hsrv\spool\` `PDVIOPDir=c:\IOPSim\`	Directory for communication with CTDOS terminals Directory for communication with the IOP

`For debug purposes only:`
 	
`InfoFenster=100`
 	
`SlowDown=600`
	
	Number of lines of the “current” window (presentation of last PDV actions)
	
	Slow down, to make events “visible”

`;Supported barcodes`	
`FieldWNRBarcodeOnly=Y`	If this entry is set the tool number may only be entered using a scanner.
`FieldNestBarcodeOnly=Y`	If this entry is set the cavity number may only be entered using a scanner.
`FieldNummBarcodeOnly=Y`	If this entry is set the number may only be entered using a scanner.
`FieldKNRBarcodeOnly=Y`	If this entry is set the badge number may only be entered using a scanner.
`BarcodeWNR=`	This field specifies which acronym is entered into the tool number field by the scanner.

Entry	Comment
`BarcodeNest=`	This field specifies which acronym is entered into the cavity number field by the scanner.
`BarcodeNumm=`	This field specifies which acronym is entered into the number field by the scanner.

6 Central Configuration File hytnrcfg.ini

This file includes different configurations for all or single terminals at a central place.

Each section is available in a generally accepted version
[section 0].

However, entries included in this section can be overwritten by entries in a terminal-specific section [section <TNR-USER>]
<TNR-USER> = HydraUser = Terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/HYDRA User

The hytnrcfg.ini file is loaded from the server every time the terminal is started.

Section / Entry	Comment
	→→→
[Tnr configuration 0]	
`FollowExternStatus=on`	Transfer of machine statuses when reloading machine list Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
`InstallFonts=on`	If this is set to "off" fonts will not be installed during the restart. ON=DEFAULT
`OnlyInstallFontsAfterDownload=false`	"`InstallFonts=on`": If true then fonts will only be installed directly after a download. If false then fonts will be installed every time the terminal is restarted. (false = DEFAULT)
`InstallTvicport=on`	If "off" the LPT driver "tvicport.sys" will not be installed. It is required for HYDRA-ZKS. ON = DEFAULT
[Terminal->USR 0]	

`AttachedApplication=First`	This configuration checks whether or not an application is connected in Windows that matches the file extension of the document to be displayed from the OP info dialog. If there is such an application, it will be used for displaying the document. If there is no connection, viewers configured in ctaip.ini (→ [ext. software]) and internal viewers will be used. In case, an extension is completely unknown it is attempted to display it as text Different settings may be configured: **First** → search for connected application first **AfterUserViewer** → If a UserViewer is configured this one overrides the connected application (also applies for ExcelViewer, WordViewer and PowerpointViewer) **Last** → Only if no ctaip.ini assignment is found for the file extension, then the connected assignment will be searched for (**default**). **Off** → Connected application is never searched.
`HTTPBrowser=standard`	Viewing of documents (via OP info): If documents are configured with a path of the type "http", the file will not be downloaded to the terminal, but the link will only be transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows will be used.
`SupressErrorMessage=70012`	Suppress message "material is not planned"
[SignatureRecording->User 0]	
`ManualBadgeInput=true`	This configuration specifies whether or not the field "user" can be edited in the terminal (by default: no editing) **true** → activates keyboard input for the "user" field in the terminal

`Transparency=255`	The signature dialog can also be transparent. **255** → Signature dialog is 0 % transparent (not transparent) **1** → Signature dialog is 99% transparent (maximum transparency) (Default = 155) Available as of CTAIP (V# 2.0.2.25) / CTWIN (V# 7.2.5.99)
`ShowPosition=TR`	The position of the signature dialog can be adjusted as follows: **TL** Top – Left **TM** Top – Middle **TR** Top – Right **ML** Middle – Left **MM** Middle – Middle (Default) **MR** Middle – Right **BL** Bottom – Left **BM** Bottom – Middle **BR** Bottom – Right Available as of CTAIP (V# 2.0.2.25)
`USE_SERVICE_ACCOUNT=1`	**0** (default) SSO: do not use ServiceAccount (requires the terminal to be started with the "user" domain (SSO). **Please note:** ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
`SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY`	**REPORTING_USER_READONLY** The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field is read-only. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. **REPORTING_USER_CHANGEABLE** The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field can be modified. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible.

`SIGNATURE_1_LOGON_TYPE=HYDRA`	**"" / Not set / "EMPTY"** There is also an alternative login procedure. **HYDRA** The tab identifying users via the Windows user is blocked. The HYDRA user must be used for identification purposes. This requires, however, that in the HYDRA HR master all users logging in are created and that the "SSO" option is not set. Otherwise, successful authentication is impossible. **ACTIVEDIRECTORY** The tab identifying users via the HYDRA user is blocked. The Windows user must be used for identification purposes. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. **MIXED_BUT_UNIQUE** Either the HYDRA or Windows login procedure is available, subject to whether or not the "SSO" option is set for the registered user in the HYDRA HR master. "SSO" enabled → Windows only "SSO" disabled → HYDRA only
`SIGNATURE_2_LOGON_TYPE=HYDRA`	`Identical to SIGNATURE_1_LOGON_TYPE (see above)`
`ExtendedSignatureRecording=true`	Used for signatures with the terminal in the area of quality data collection.

6.1 Layout configuration

Entry	Comment
Section and/or **[terminal configuration 0]** **[terminal configuration 2XXX];**	**(general configuration)** **(2XXX terminal-specific configuration)**
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE	In case of an error in reading the clock (e.g. after coming out of standby mode), this configuration makes sure that the time is accepted without having to confirm a dialog. Afterwards the terminal time will be synchronized with the server time using a PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON	As of ctaip V# 2.0.2.23 Prevents the warning after restarting the terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default = OFF)

Entry	Comment
NetRuntimeMode=2	As of ctaip V# 2.0.2.50: Alternative calculation of the target quantity since logon: The net run time is not calculated from the times when the production lock is enabled (PSperre=green) but only from the shift times less the shift breaks. Consequently, it can also be displayed, even if the terminal program has been restarted.
Section [QRD-PRINTER->TICKET 0] ;**(general configuration)** [QRD-PRINTER->TICKET 2xxx] ;**(2XXX configuration for a specific terminal)**	
COMPLETE-ABSENCE-OF-LOCAL-MNR-DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured **<Events>**, if it is not available locally => This configuration might be required/necessary for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR-DATA-FOR-EVENT=< Events >	Reloads the order row for the configured **<Events>**, if it is not available locally ➔ This option has been implemented to access order data within the master data, e.g. when logging an order on.
COMPLETE-..-EVENT=< Events > **COMPLETE-..-EVENT=#ALL#** **COMPLETE-..-EVENT=A_AN|A_P_AN**	Explanation on the configuration of **<Events>** ➔ Using **<#ALL#>** the row (ANR/MNR) that is not available is reloaded for any event. ➔ **<A_AN|A_P_AN>** restricts reloading of information to the specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.

7 Order Postings for Operations Subject to Batch Management

Summary

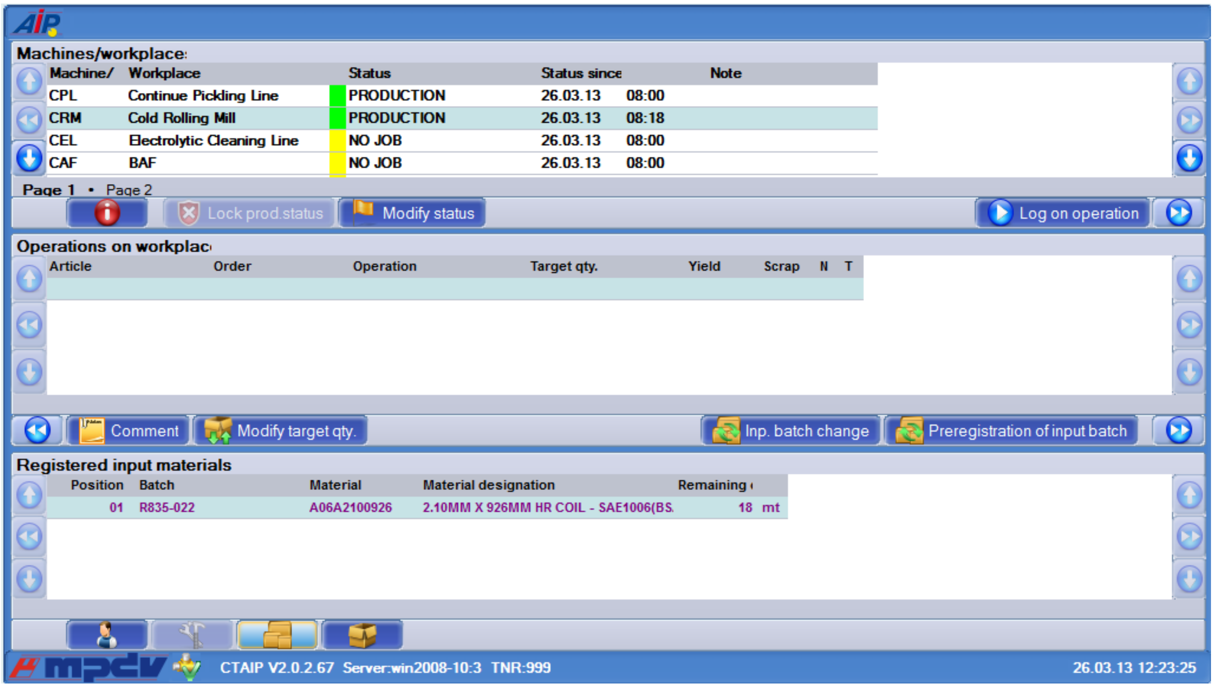
In addition to logging the actual operation on, operations subject to management in batches also enable to log on relevant input batches.

Configuration

These system settings have to be made to be able to generally use operations that are subject to management in batches.

Basic screen

Basic terminal screen when a machine is assigned in batch mode:



The basic display shows the third list "Input materials currently logged on" for machines for which the “batch management” option is configured. This list shows all active input batches of the selected machine.

OP logon with input batches

By clicking the "log OP on" button a workflow including two tabs is opened. The operation to be logged on is selected in the first tab “select operation".

The “log operation on” tab is reached by clicking the “Next” button where in addition to the selected OP, the defined material components are displayed in a list.

By entering a batch number in the "input batch" field and clicking the "report batch" function a batch may be logged on as input material for a component. During the entry process, the terminal checks whether or not the batch number is known in the system and may be logged on. This is also described in detail within the document dealing with the input batch change.

"Batch" field

When an OP is logged on, a batch number is created simultaneously for the next output batch to be produced. The batch number may be assigned automatically or manually (please also see the settings of "workplace configuration"). The generated batch is created with the batch number in the system and set to the "running" status.

Provided that all required input materials are logged on, the OP may be started via the "OK" button in the "OP logon" dialog. Whether input material has to be logged on or not, can be defined in the assigned material type of the component.

Once the OP has been logged on successfully, all active input batches of the selected machine are displayed in the material list.

If batches are logged on along with the OP and the user cancels the process or cannot log the OP on due to validation checking the input batches will be logged off automatically for this OP. In this case, batches are always logged off without indicating the consumption quantity. By way of the following warning message, the user may confirm the logoff process:

The function that logs input batches off automatically can be activated/deactivated by an option in the hyaipcfg.ini file

HYAIPCFG.INI
[MPL-Options 0/2xxx] ForceAutoLogOffInputBatches=0

Logon of unplanned input material

In addition to planned materials, it is also possible to log “unplanned” material on for an OP, using an additional feature at the component of the OP. If the "replaceable" option is set to "J" the user is able to assign the respective component manually, when batches are logged on. However, the logon is only allowed if the material type of the input batch corresponds to that of the component.

Within the selection list the components are filtered according to the material number and displayed as follows:

Logon of unknown batches

An input batch, which is not yet known in the system, may be logged on for an OP using the "create unknown batches" option in the basic parameter settings.

In this case, it is searched for a valid assignment of input material to the material type of the selected component, when input batches are logged on. Provided that a corresponding assignment is found and the "allow entry of unknown input batches" option is configured in the "input batch processing" tab at the material type, the batch is generated by logging it on to the system and set to the "running" status at the same time. The batch is initially created in a quantity of 1.000.000.000.

Logoff/interruption of OPs

A running OP may be interrupted or logged off by clicking the "logoff/interrupt OP" button. Then a dialog opens, where the following selection can be made:

If "log OP off" is clicked the logoff dialog opens that contains the same input fields like the output batch change dialog.

Thus, the output batch that is currently still active is completed, when OPs are interrupted or logged off.

8 Input Batch Change

Summary

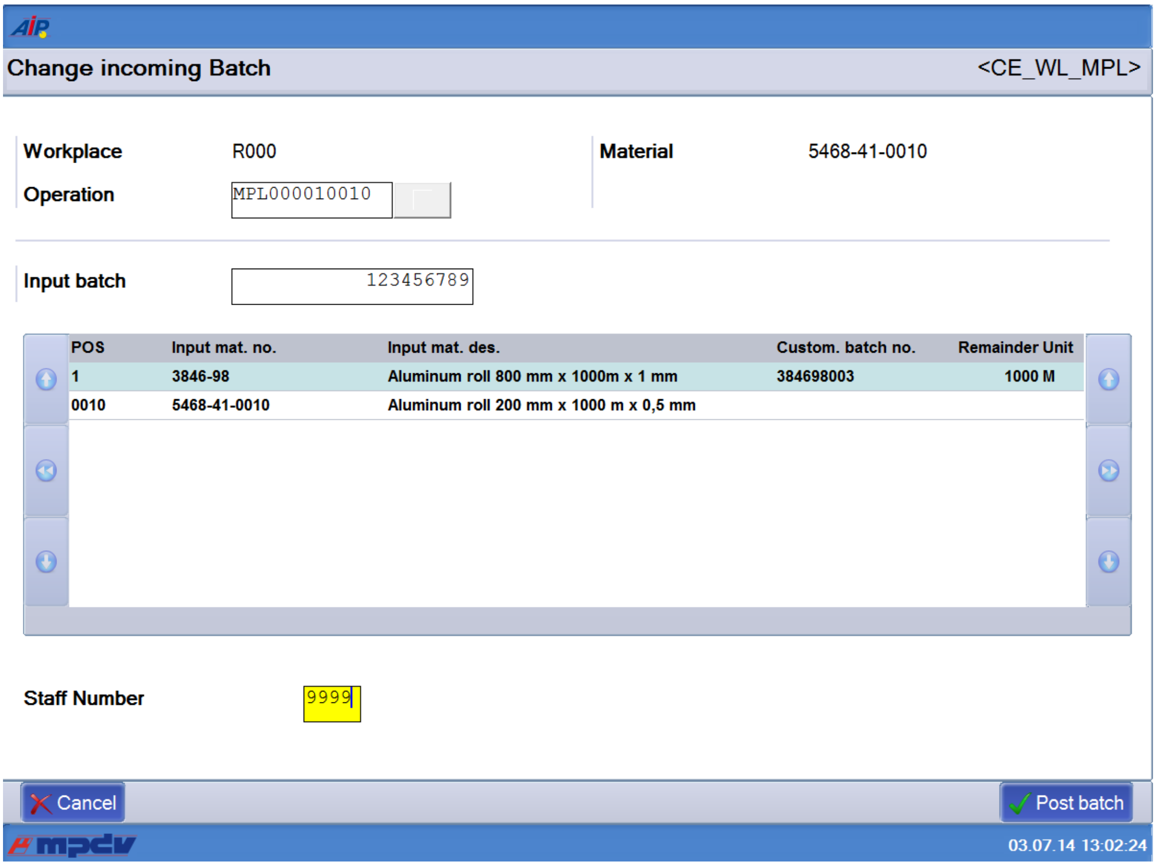
Input material and/or relevant input batches can be changed for a running OP if the "input batch change" option is clicked.

Configuration

Further system configurations are not required to be able to use the input batch change dialog.

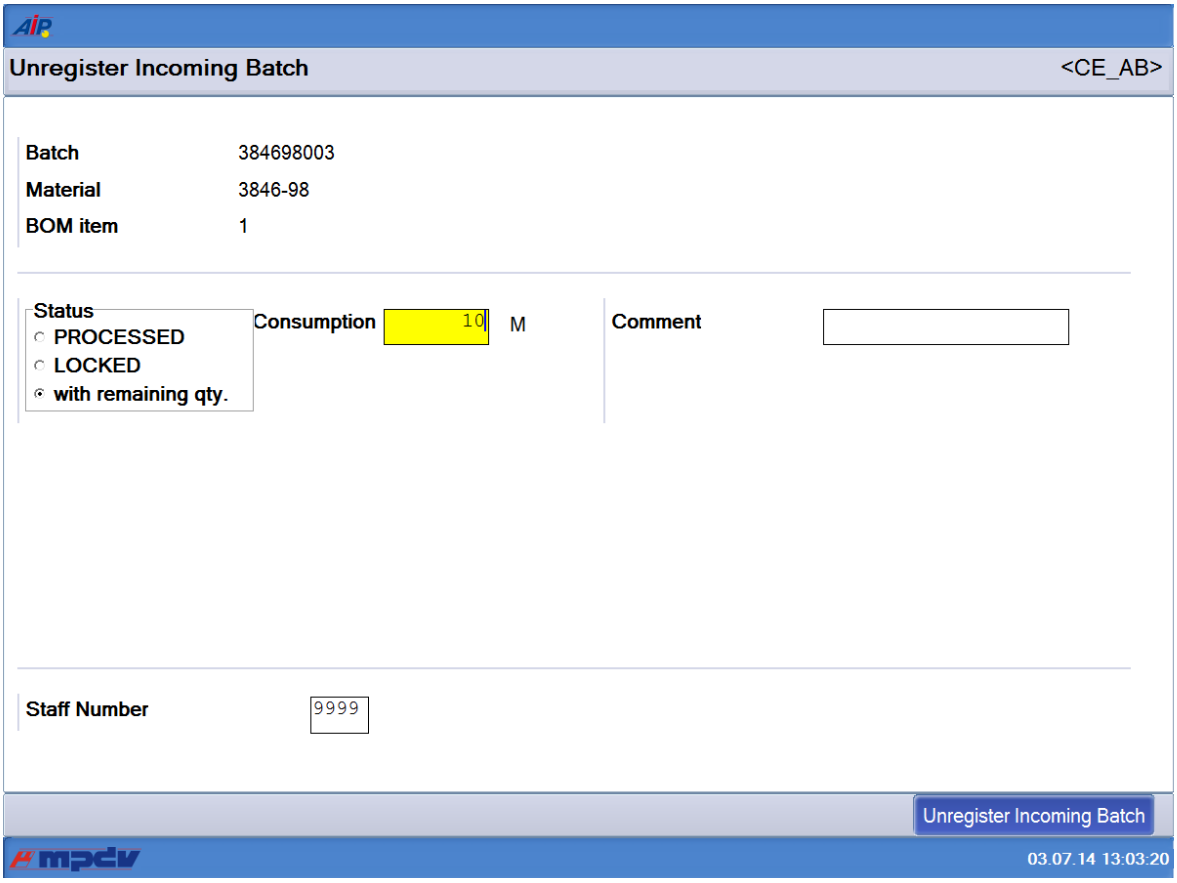
Dialog

Basic screen



Log input batch off:

Input batches may be changed by entering a currently active batch number or by entering a new batch number. When logging batches off, it is also possible to enter the status and consumption of the batch to be logged off.



Options when logging input batches off:

F1 - PROCESSED

The batch is set to the "processed" status and the remaining quantity that is still available is set to 0. A consumption posting is generated as goods issue for the current, remaining quantity.

F2 - BLOCKED

The batch is set to the "blocked" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue.

F3 - with remaining quantity

The batch is set to the "free" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue. If the remaining quantity that is still available becomes <= 0, the batch status automatically switches to "processed".

Consumption

The entered consumption (unit of the input material) is deducted from the remainder of the batch and a goods movement is generated.

Comment on batch

The comment entered is saved as information for the batch.

Log input batch on:

Provided that the batch is known, batch data is displayed in an intermediate dialog where the logon may be confirmed.

Provided that the batch could be logged on, it is taken over to the material list in "customer batch number" and thus the change is completed.

However, in case the logon is inadmissible as the input material does not correspond to that of the component, the logon is rejected by an error message.

9 Output Batch Change

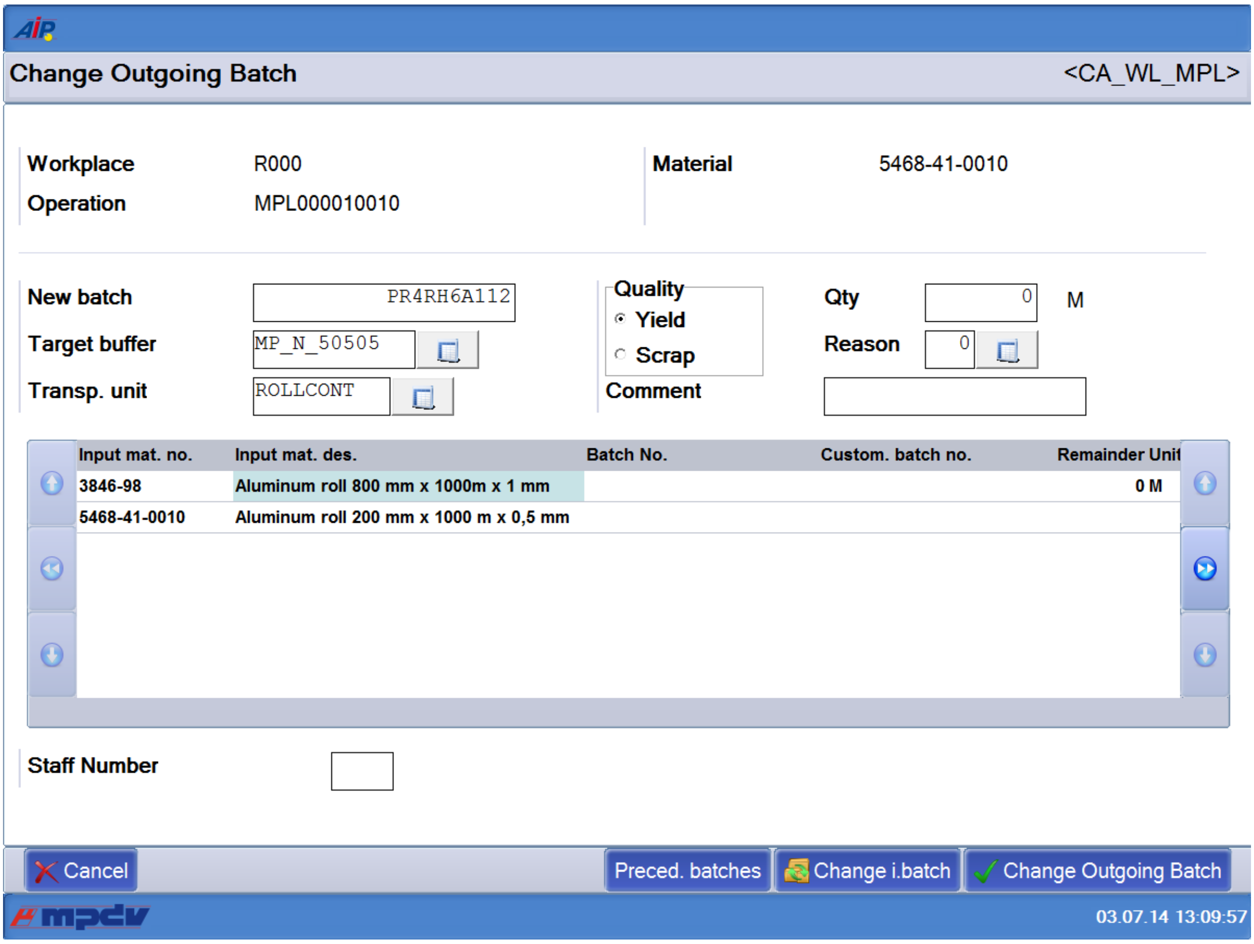
Summary

Output material may be changed for a running OP using the "output batch change" option.

Configuration

Further system configurations are not required to be able to use the output batch change dialog.

Dialog



The input batches that are logged on are displayed with their available remaining quantity in a list within the output batch change dialog. The following data may be entered:

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here. The selection refers to transport units that were assigned to the material type of the OP.

Comment on batch

In this field a comment may be saved for the batch to be produced.

Quantity

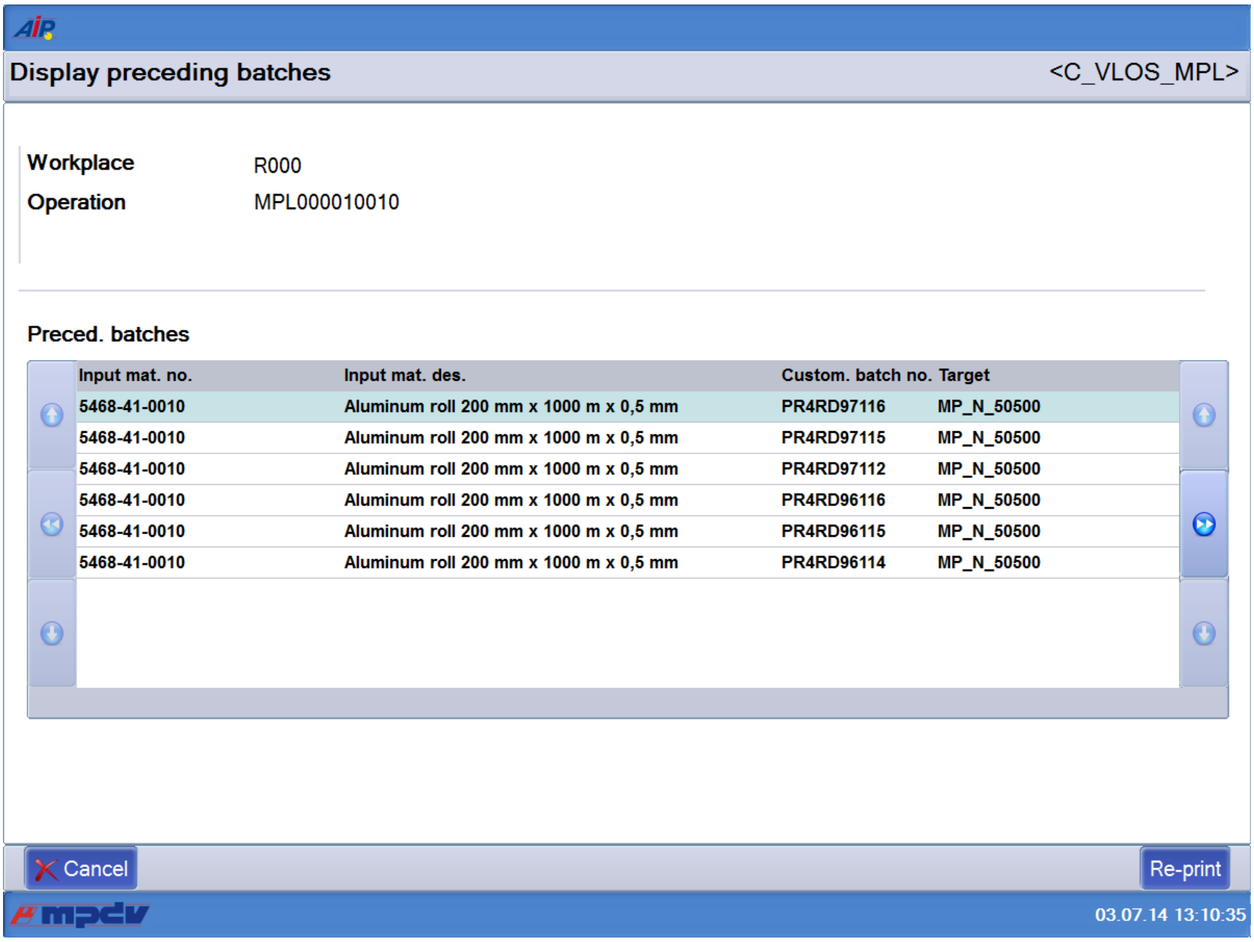
The batch to be produced is posted with the quantity entered here. The quantity is taken over as primary quantity of entry to the order and machine and a goods movement is generated as goods receipt.

Quality

A batch may be classified as yield or scrap quantity. The system posts yield batches with the "free" status. Scrap batches automatically get the "blocked" status. When scrap is selected, a valid scrap reason has to be assigned.

"Preceding batches" function key:

This button opens a list with output batches which have already been produced for this OP.



"Change inp. batch" function key:

Using this button the user can switch to the "input batch change" function.

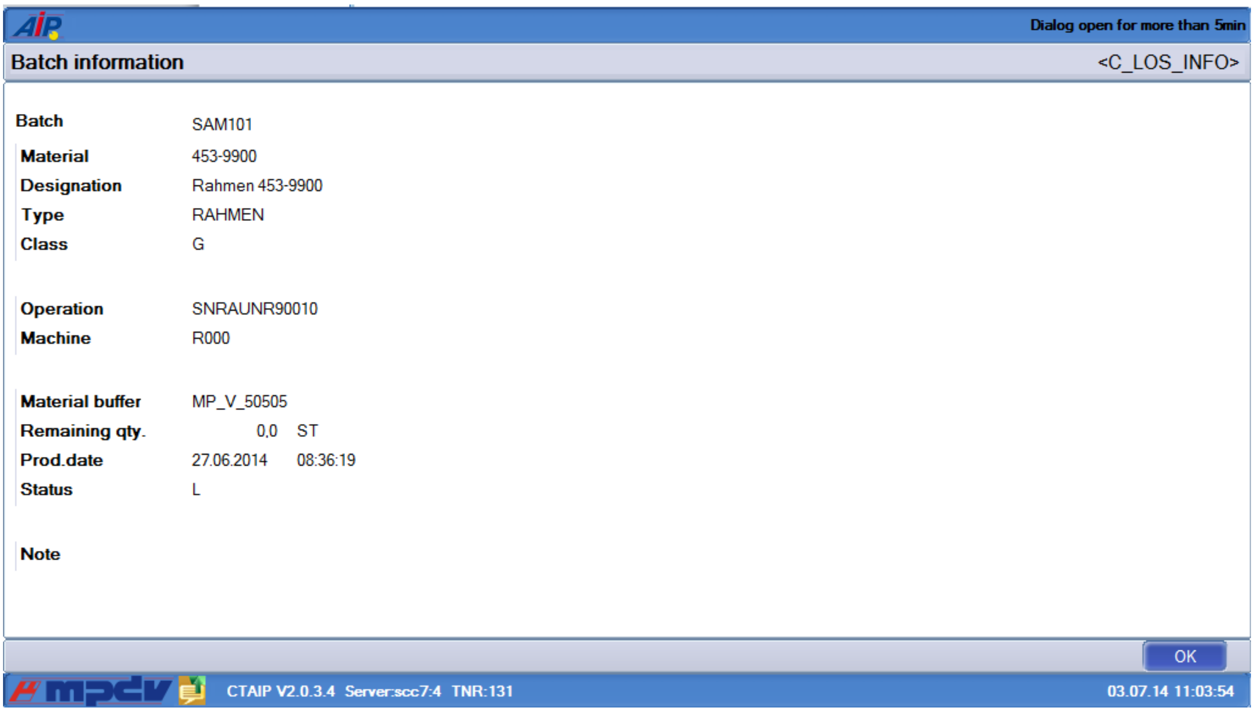
New batch

When a current output batch is completed, a new batch number is simultaneously created for the next batch. The batch number may be assigned automatically or manually. The batch generated in this way is created with the batch number in the system and set to the "running" status.

10 Batch Information

Batch information

Batch information is displayed in a dialog, when the “batch info” icon is clicked.



11 Entry of Batch Attributes

Summary

In case batch attributes are defined to be recorded at the AIP terminal for the material type of the running OP, another input dialog is opened, when the output batch is changed. The dialog opens additionally after clicking "OK" in the output batch change function, interrupt OP and finish OP function.

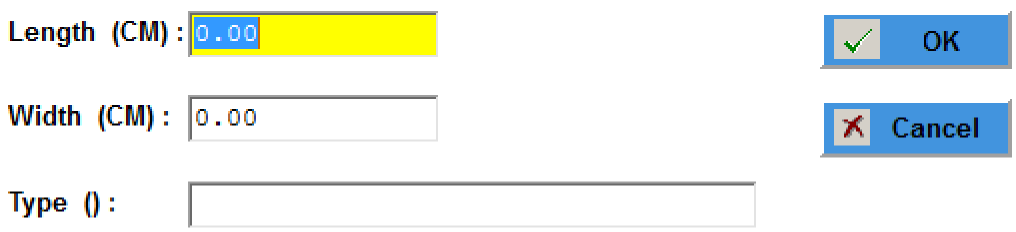
Using batch attributes, numeric and alphanumeric values may be recorded which are then saved for the produced output batch in an additional table.

Configuration

By configuration it is possible to record any number of additionally required batch attributes for the material type.

Dialog

Example when two additional batch attributes are collected for an output batch:



12 Enter Goods Receipt Batch

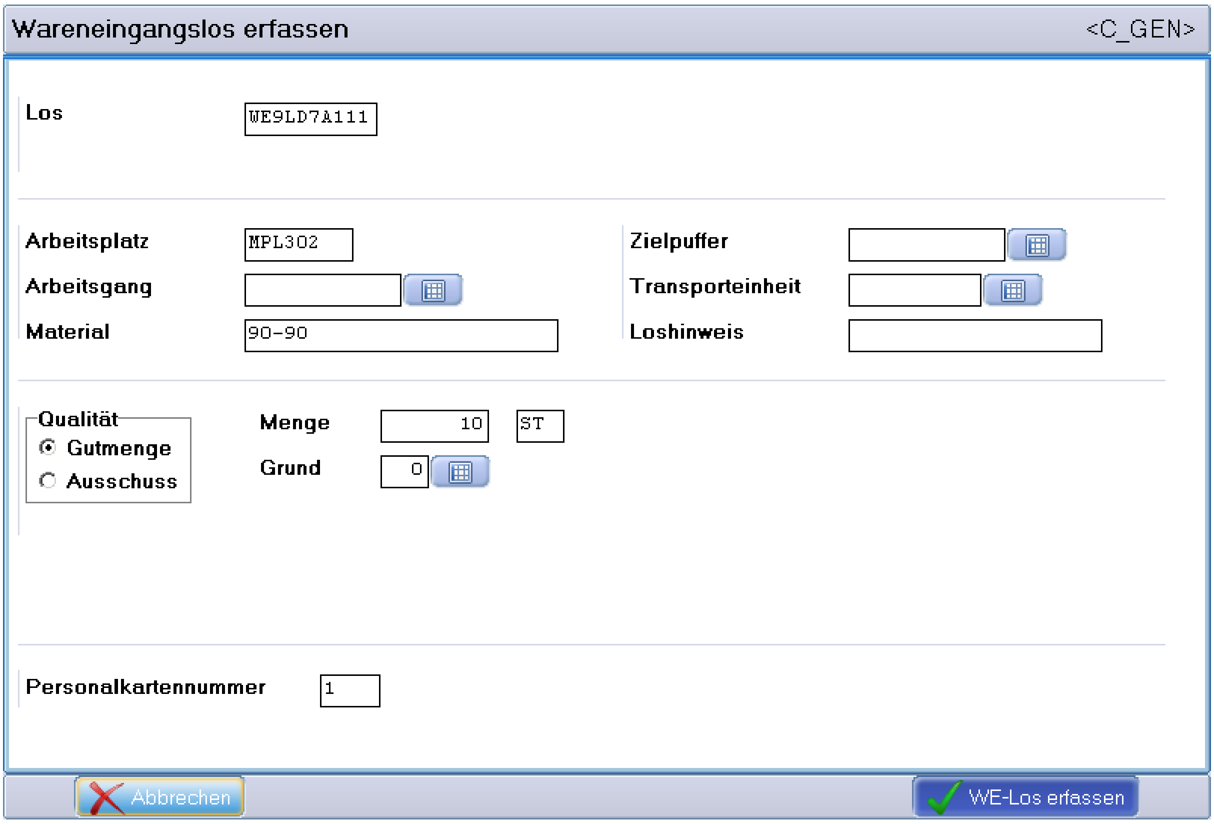
Summary

A new goods receipt batch may be created in the system via the "enter GR batch" button.

Configuration

Further system configurations are not required to enter goods receipt batches.

Dialog



Having clicked the "OK" button, the batch is created and the dialog remains open for further entries.

Batch numbers may be generated automatically or manually depending on the configuration. Moreover, the following data is saved at the batch.

Workplace

Machine where the batch was recorded

Operation

Order where the batch was recorded

Material

Material number of the batch

Quantity and unit

The batch is created with the quantity and unit entered here. A goods receipt is posted with the quantity.

Quality

A batch may be classified as yield or scrap quantity, rework or open quantity. The system creates yield batches with the "free" status. Scrap batches automatically get the "blocked" status and the batch class "scrap". When scrap is selected, a valid scrap reason has to be assigned additionally with respect to the corresponding workplace.

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here.

Comment on batch

In this field a comment may be saved for the goods receipt batch.

13 Repost Batch

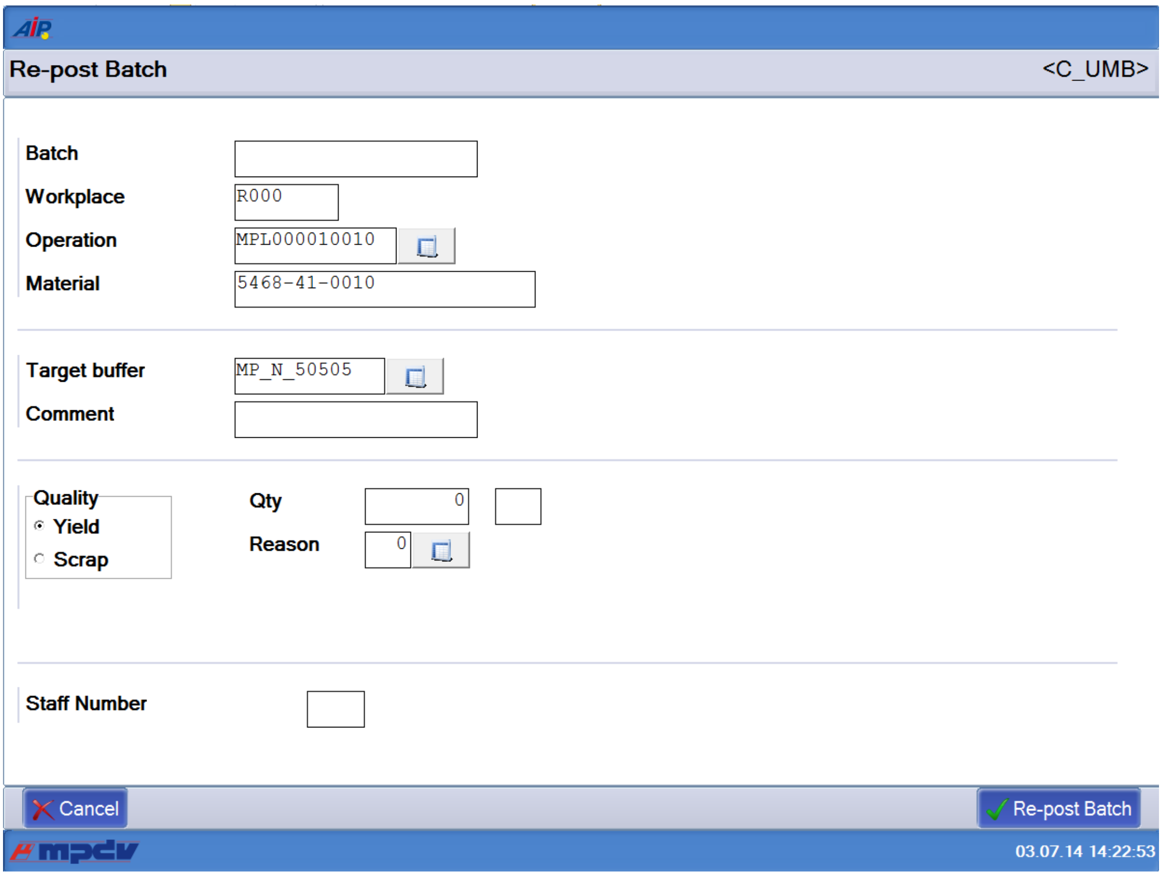
Summary

Using the "repost batch" button, an existing batch may be reposted to another material buffer.

Configuration

Further system configurations are not required to repost batches.

Dialog



Having clicked the "OK" button, the batch is reposted to a new material buffer and the dialog remains open for further entries.

As an alternative, the batch can also be reposted from yield to scrap.

14 Display of "Produced Batches"

Summary

Using the third list, it is possible to display the output batches produced for a running operation that is subject to batch management within the machine master at the terminal.

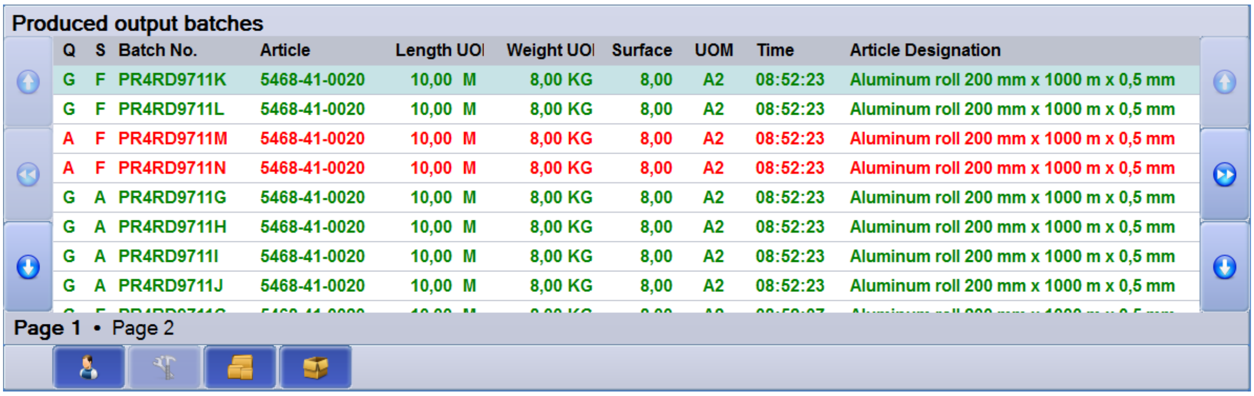
A default number of 20 output batches is displayed for each machine in the list. Output batches (yield and scrap batches) are shown, which have been produced at this terminal since output batches were changed.

As this list is only kept locally by the respective terminal, it is not synchronized with the server, when AIP is started.

Configuration

The settings required to use the third list can be found in the document dealing with the configuration.

Display


Produced output batches

Q	S	Batch No.	Article	Length UOI		Weight UOI	Surface	UOM	Time	Article Designation
G	F	PR4RD9711K	5468-41-0020	10,00	M	8,00 KG	8,00	A2	08:52:23	Aluminum roll 200 mm x 1000 m x 0,5 mm
G	F	PR4RD9711L	5468-41-0020	10,00	M	8,00 KG	8,00	A2	08:52:23	Aluminum roll 200 mm x 1000 m x 0,5 mm
A	F	PR4RD9711M	5468-41-0020	10,00	M	8,00 KG	8,00	A2	08:52:23	Aluminum roll 200 mm x 1000 m x 0,5 mm
A	F	PR4RD9711N	5468-41-0020	10,00	M	8,00 KG	8,00	A2	08:52:23	Aluminum roll 200 mm x 1000 m x 0,5 mm
G	A	PR4RD9711G	5468-41-0020	10,00	M	8,00 KG	8,00	A2	08:52:23	Aluminum roll 200 mm x 1000 m x 0,5 mm
G	A	PR4RD9711H	5468-41-0020	10,00	M	8,00 KG	8,00	A2	08:52:23	Aluminum roll 200 mm x 1000 m x 0,5 mm
G	A	PR4RD9711I	5468-41-0020	10,00	M	8,00 KG	8,00	A2	08:52:23	Aluminum roll 200 mm x 1000 m x 0,5 mm
G	A	PR4RD9711J	5468-41-0020	10,00	M	8,00 KG	8,00	A2	08:52:23	Aluminum roll 200 mm x 1000 m x 0,5 mm

Page 1 • Page 2

Figure: Display of the output batches produced

Using the  icon, the user can switch to the display of output batches.

The list includes, among other things, the article, article designation, batch number, date, time, quantity and batch class, user fields as well as alternative batch numbers. Some of the fields might not be assigned values depending on the entry scenario when output batches are generated.

15 Settings for List of Produced Output Batches

Activation at the machine

For the purpose of activating the list of produced output batches in the basic screen of the terminal, the display of the third list of the machine to be activated has to be activated in the machine master record (Workplace configuration → Entry → Display 3rd list).

Definition of the Number of Entries in the List

The number of entries in the list may be customized by extending the data provision for the machine list (mnr.lst) at the terminal on the server side (extended customizing). For this purpose, the additional column "MNR.NUMBER_OF_BATCHES" has to be provided at the server.

Customization of Visible Columns in the List

The list contents can be configured in section [MNR_AMAT.LST] of the file ctaiplay.ini.

Example:

CTAIPLAY.INI

```
[ MNR_AMAT.LST ]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=ROW.IDX=-
GRID_CAPTION=Produzierte Ausgangslose

EXAMINE_SCANEXPR1=KLASSE=G
EXAMINE_SCANCOLOR1=clGreen
EXAMINE_SCANEXPR2=KLASSE=A
EXAMINE_SCANCOLOR2=clRed

; ROW.IDX=N10,50,R,Row
CNR=C20,150,L,Losnummer
;CNR=*CNR,Los
ATK=C25,125,L,Artikel
; ATK=*ATK
; KLASSE=C3,40,Z,*
MENGE=N12.0,70,R,Menge
EINH=C3,30,Z,ME
DAT=dd.mm.yyyy,70,L,Datum
ZEI=hh:mm:ss,60,L,Zeit
ATKBEZ=C30,200,L,Artikelbezeichnung
```

Configuration of Server-Based Comparison

The server-based comparison is activated by an entry in the customer-specific configuration file "*ctwinlisten.ini*". As with hytnrcfg.ini, this file can be maintained both globally (for all terminals), on a terminal group level, and on terminal level.

Activation is effected by entering a LOADCYCLE larger than 0 (seconds).

CTLISTEN.INI
`[#LIST#TNR-ALOSE]` `LOADCYCLE=900`

The prerequisite for the server comparison is the default configuration file "ctlisten.cfg", which contains the application-specific configurations of the server list.

CTLISTEN.CFG

`[#LIST#TNR-ALOSE]`
`CMD=DLG=LIST;13|MOD=P|     ; (Default <ANZ=250>)`
`LOADCYCLE=0`
`QUEUEEMPTY=TRUE`
`FORCENOTIFY=TRUE`

16 Advance Logon of Input Batches

Summary

The process might require an input batch to be logged on in advance and set up accordingly on a machine, while the preceding input batch is still being used for a material.

This situation frequently occurs at very large machines processing, for example, roles or belts that are uncoiled as input batch at the beginning of the machine and coiled up as output batch at the end of the machine.

As the users are mostly busy with activities at the end of the machine at the time when the input batch actually needs to be changed, they cannot perform the input batch change and, as a result, they are provided with the opportunity to log the next input batch on already in advance for an order/OP.

Then the input batch can actually be changed by logging a new OP on or a project-specific call can be established.

General / usage

The function for logging input batches on in advance is used to be able to "set up" and "log on" the next input batch while an OP and input batch are still running. This next input batch is not yet running but assigned the "logged on in advance" flag.

An input batch may be logged on in advance for a currently running OP or a prepared OP.

Configuration

The settings required for using the function “advance logon of input batches” is described here.

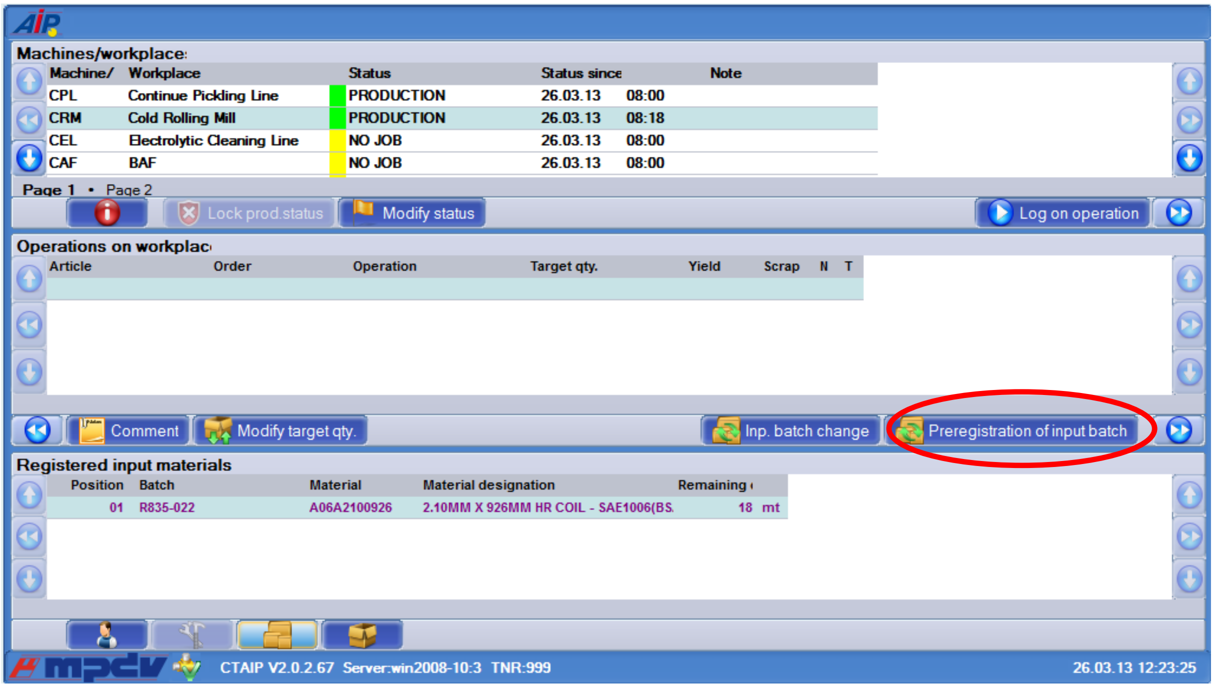
Procedure

The procedure for using the function “advance logon of input batches“ or the logical process is described here.

Dialog

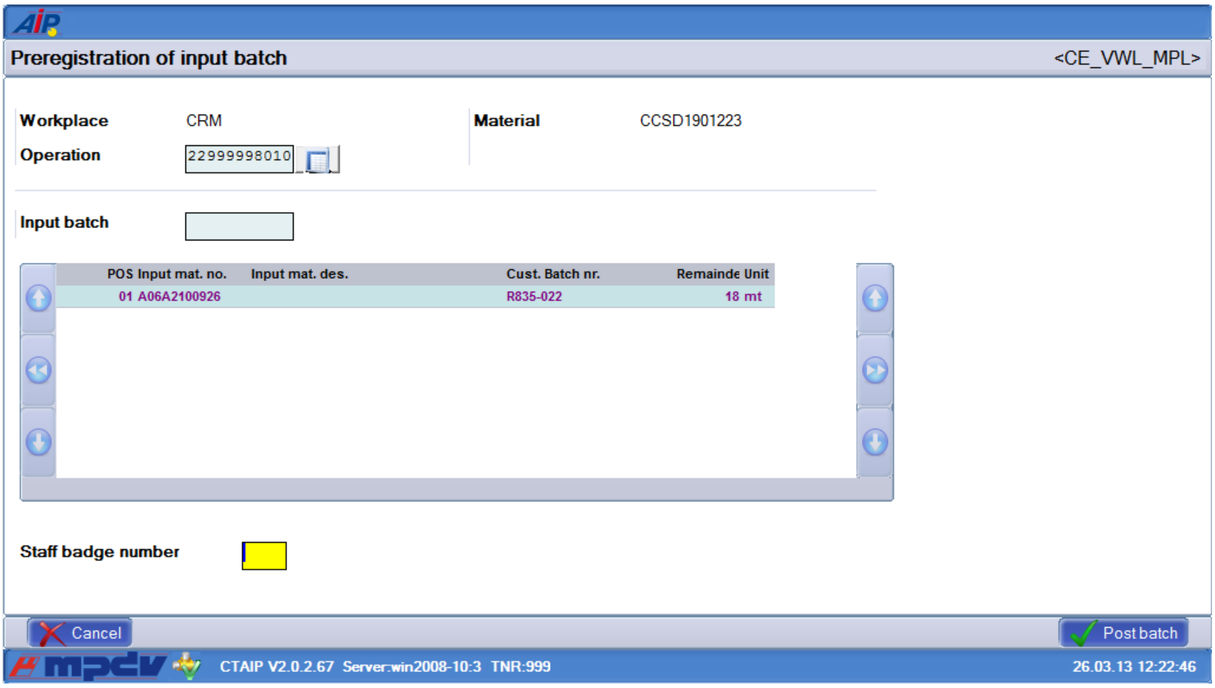
Basic screen

The basic AIP screen shows the function key “Advance logon of input batch” (preregistration of input batch). The dialog for logging input batches on in advance may be used by clicking this function key.



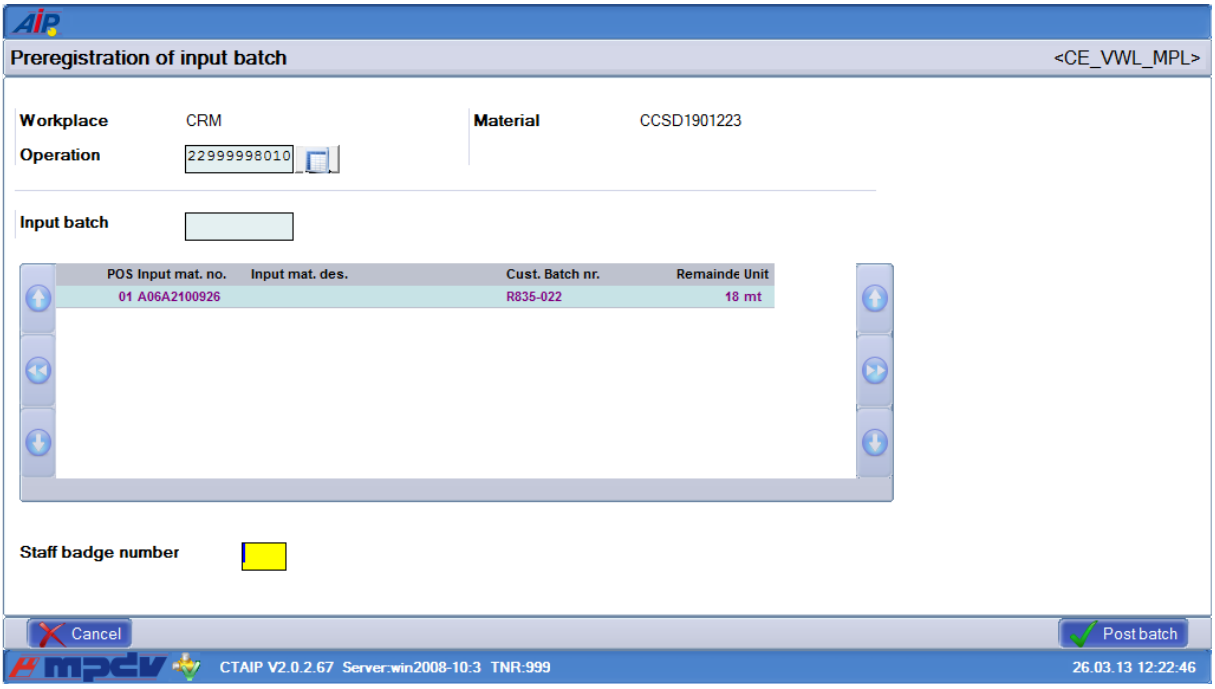
Advance logon of input batches (CE_VWL_MPL)

The user selects the workplace to which an input batch is to be logged on in advance in the basic screen. The below dialog (CE_VWL_MPL) opens by clicking the function key ”Advance logon of input batches” (preregistration of input batch).



If an operation is currently running/logged on to the workplace, this one will be selected by default. The input batch (that is to be logged on in advance) is entered/scanned for the selected BOM item. Advance logon of input batches is started by clicking the button “post batch".

At first the input batch is checked for validity (dialog CE_VAN). The material number of the input batch is checked against the material number of the component list or the BOM item. The input batch is logged on in advance, once the button “log input batch on in advance” has been clicked:



Finally, the input batch that has been logged on in advance is displayed in purple in the BOM of the component.

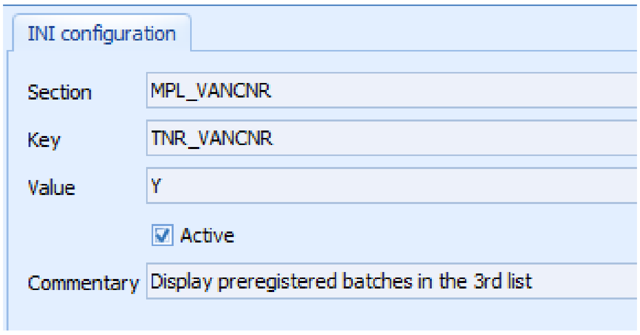
The dialog can be closed with the “cancel” key.

17 Configuration for Preregistered Input Batches

Configuration for Display of Third List on AIP (INI Configuration)

Please set the following parameters/values in the INI configuration for displaying input batches logged on in advance in the third list on AIP.

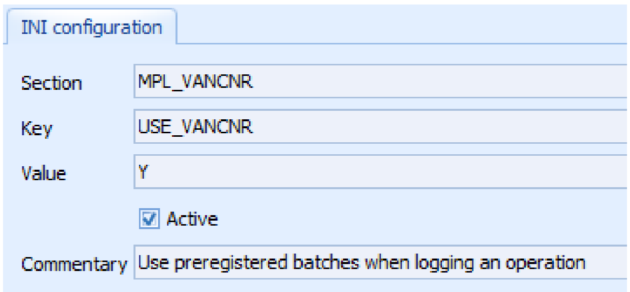
- Ini name: MPL
- Section: MPL_VANCNR
- Key/Value: TNR_VANCNR =Y



Configuration for Logging on Input Batches when Logging On an OP on AIP (INI configuration)

Please set the following parameters/values in the INI configuration so that the input batches logged on in advance are also logged on/considered when an OP is logged on to AIP.

- Ini name: MPL
- Section: MPL_VANCNR
- Key/Value: USE_VANCNR =Y



Configuration for Keyboard Layout on AIP (Ctaipbut.INI)

For the function key display on AIP in the basic view, the following entry must be made in the configuration file Ctaipbut.ini. The file is to be saved accordingly on the server.

Entry in Ctaipbut.ini:

```
[ANR-LN-Page2]

1=A_INFO.Dialog1,L,BDE-Kommentar,Attach Notes.png

2=A_SMG,L,Sollmenge ändern,Shipping Box Open Move Down Up.png

3=A_ELW,R,Eingangsloswechsel,CE_WL.png

4=CE_VWL_MPL,R,Eingangslosvoranmeldung,CE_WL.png

5=%BART:CAQ=J%CAQ_DC_T,R,Prüfung durchführen,Generators.png
```

Configuration for Marking Input Batches Logged On in Advance on AIP (Ctaiplay.INI)

To enable color marking of the input batches logged on in advance in the input batch list, material list and BOM on AIP, the following entry must be made in the configuration file Ctaiplay.ini. The file is to be saved accordingly on the server.

Entry in Ctaiplay.ini:

`Coloring of preregistered batches:`

`Sections [input batch list], [Material list]` and `[ FHM list (KOMBI) ]:`

```
...
EXAMINE_SCANEXPR1=CST=X
EXAMINE_SCANCOLOR1=ClPurple
...
```

18 Throughput Batch Processing

Overview

Throughput batch processing is a special application case in material and production logistics. In this case, an output batch is to adopt the visible batch number of the input batch used.

Usage/Procedure

In contrast to normal input and output batch processing in an operation requiring batch management, in throughput batch processing an input batch with a batch number is used and this batch number is handed down to the output batch.

Throughput batch processing is used for instance if the condition of a batch/material might change after a work stage but the external batch number (e.g. on a label) is to be retained.

As a consequence, no new material will occur when the output batch is changed and the batch number can even be handed down through several process steps.

The operator uses an input batch on a machine/operation. The output batch number is not changed and hence remains identical to the input batch number.

The so-called throughput batch number (throughput batch number/external batch number) therefore remains identical. Within the system, however, a unique batch number (HYDRA batch number/internal batch number) is still used and/or generated for each production level/after each output batch change, since every object within the system is unique.

If, for instance, a selection according to throughput batch numbers is made in the batch data overview, several entries with different internal batch numbers will be obtained for each throughput batch number; these internal batch numbers ensure unambiguity within the system and consequently allow for a historic observation of the "throughput batch".

This means that a total of three different statuses are considered for the process description:

- Status 1: The batch as an input batch (prior to logon)
- Status 2: The batch status on the machine (running on OP)
- Status 3: The batch as an output batch (after logoff)

The batch as an input batch (prior to logon):

Prior to logging on the batch, the information on the batch (e.g. in the batch data overview) reads as follows:

- the throughput batch number *dllosnr* "**DLLOS01**" is identical to the batch number *losnr* "**DLLOS01**"
- the batch status is "**F**" (free)
- the throughput batch flag *dll_kennz* is "**N**"

The batch status on the machine (running on OP)

When an operation is logged on to the terminal, the material (the material number) is used according to the component list from the OP in order to determine whether it is treated as a throughput batch on the basis of the material type entered for this material. This procedure also enables logging on the throughput batch number on this operation. It is not possible to log on more than one input batch/material as "throughput batches".

After logging on the batch (with throughput batch number/external batch number) on the operation/machine, the information on the batch reads as follows:

- the batch status is changed to "**L**" (running)
- the batch receives the throughput batch flag *dll_kennz* "**E**" (throughput input batch running)

After logging off the output batch on the OP/machine, a new internal object is created in the system to take over the throughput batch number of the input batch.

The information on the new output batch then reads as follows:

- a new batch with *losnr* **PR41E9C114** and *dllosnr* **DLLOS01** is created in the output buffer
- the status of the new batch is changed to *"***L***"* and
- the batch receives the throughput batch flag *"***G***"*.
- The quantity on the new batch is identical to the quantity of the original input batch.

Functionality: Console Evaluation of Throughput Batches

An operator can identify and trace a created batch via the external batch number (e.g. in order to be able to forecast when a specific batch will leave production).

Material Movement Functionality

An operator can call up the material movements (goods issue/goods receipt) for an input batch/output batch.

Batch History Functionality

An operator can list the batch history of an output product in order to be able to trace the manufacturing process of a batch for analyses.

Batch Tracing Functionality

An operator can use the batch tracing functionality to verify through which machines/operations the throughput batch was produced. The operator thus sees the material's route through production.

19 Throughput Batch Processing

Summary

A machine/workplace may also be configured in "throughput batch mode".

In throughput batch mode input material is continued being processed with unchanged number (throughput batch number) using an OP.

Please note: At machines with “throughput batch mode” it is impossible to log operations on at the same time.

Configuration

How to configure throughput batch processing is described here.

Dialog

The entry functions for throughput batch processing at the terminal are identical to those for active batch tracing at the machine.

20 Configuration for Throughput Batch Processing

Overview

Usage

In material and production logistics, the batch number of an input batch can be handed down to an output batch and thus make so-called throughput batch processing possible.

However, in order to enable throughput batch processing, the following configurations are required on various objects.

Machine Configuration

In the machine configuration (Master data → Workplaces/machines → Workplace configuration), the batch management in the Workplace configuration (MPL) tab is to be set to value "**D**".

When throughput batch processing is active on the machine, the "Automatic generation of batch number" option cannot be used, since in this case the batch number will always be handed down by the input batch.

The entry of machine cycles in combination with throughput batch processing is not used in general, since there is always a 1:1 transfer of input batches into output batches. For this reason, throughput batch recording is generally only used in connection with manual unit posting at the terminal (e.g. use in furnace, conditioning, etc.).

Configuration of Material Type

In the master data configuration (Master data → Material → Material type), the material type is to be configured in such a manner that the batch number is "handed down" and the input batch is only valid for one output batch.

- Retrograde inventory management is generally not performed for the component whose batch number is transferred as a throughput batch, since consumption is always 1:1.
- A parallel log-on of the input batch on several machines is not supported by the system.
- The entry of unknown input batches is not supported by the system.

Configuration at the Operation

The configured material type is to be entered as the material type at the operation.

The operation is to be identified as requiring batch management.

Configuration at the Operation - Component

The configured material type is to be selected as the material type at the component.

In addition to the component for which the batch number is to be handed down, other material components can be maintained at the operation. These continue to be taken into account in the usual way in the course of batch log-on and consumption recording.

21 Batch Consumption

Usage

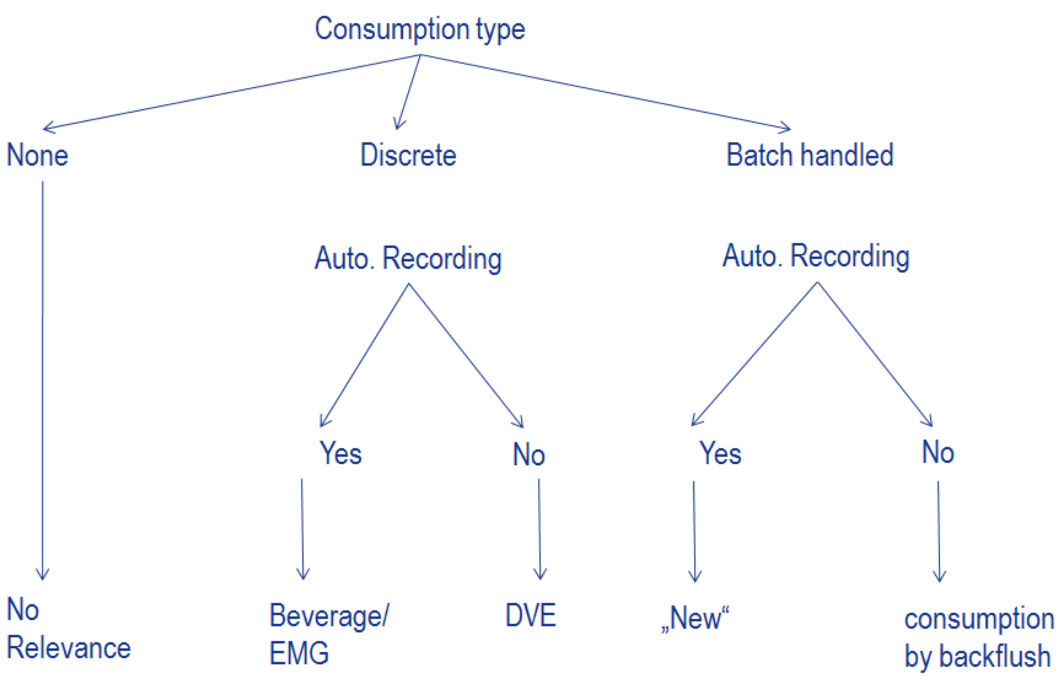
Material is used and represented in the system by:

- material postings to regulate inventories with/without ERP and without tracing
- batch-related material postings to trace back the recorded parts/materials

Subject to the type in use, consumption can be recorded differently in the system.

Procedure

These types of consumption recording are used in MES:



<u>Discrete consumption recording:</u>

Discrete consumption recording is used every time when a discrete amount of consumption can be entered for the used components by the user or a counter. Data is only entered for the component/material number. Data may be collected:

- automatically (configuration of a consumption meter per material)
- manually (the user enters material consumption manually)

<u>Batch-related consumption recording:</u>

Batch-related consumption recording is used every time when batches are used for the components for which a consumed quantity can be entered by the user or a meter. Data is collected regarding the input batch. Data may be collected:

- manually (dialog to log off the input batch and to enter batch consumption manually)
- in a retrograde manner/backflush (automatic calculation of batch consumption by generating the output batch quantity)
- automatically (automatic collection of batch consumption by a consumption meter)

Discrete consumption - manual

How to enter discrete consumption is described here.

Discrete consumption - automatic

General

Automatically recorded consumption is collected by a meter configured at the machine. Data is collected for a material type of materials included in the component list.

Configuration

These configurations have to be set in the system if material consumed discretely is to be indicated/counted by a meter:

- Component of the OP:
  The "consumption type" has to be set to "D = discrete".
- Material type of the material:
  The option "inventory management" has to be set to "N = No".
- Meter for the material type of the material:
  Configure meter like MDE meters.
  Option "compensation with material" = yes
  Option "material type" = material type of the material that is consumed

Posting/result

A goods issue is generated for automatically recorded consumption in the system.

Manual batch consumption

General

Manual batch consumption is entered by the input batch change function. The user enters the consumed quantity when logging the used input batch off.

Configuration

These configurations have to be set in the system if material is to be consumed manually as input batch:

- Component of the OP:
  The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
  The option "inventory management" has to be set to "E = Yes, when logging input batch off".

Posting/result

The consumed quantity is deducted from the remaining quantity of the input batch and the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.

How to enter batch-related consumption is described here.

Retrograde batch consumption

General

Retrograde batch consumption is calculated continuously as the output batch quantity increases. When logging the input batch off, the remaining quantity of the input batch is reduced by the calculated consumption quantity. Usually, the user does no longer enter a quantity when logging the used input batch off.

Configuration

These configurations have to be set in the system if material is to be consumed in a retrograde manner as input batch:

- Component of the OP:
  The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
  The option "inventory management" has to be set to "R = Yes, backflush (retrograde)" or "G = Yes, backflush (only with YIELD batch), retrograde".

Posting/result

The quantity calculated in a retrograde manner is deducted from the remaining quantity of the input batch and then the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.

Automatic batch consumption

General

The automatically recorded batch consumption is collected continuously as the meter quantity increases. When logging the input batch off, the remaining quantity of the input batch is reduced by the automatically recorded consumption quantity. Usually, the user does no longer enter a quantity when logging the used input batch off.

Configuration

- Component of the OP:
  The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
  The option "inventory management" has to be set to "R = yes, backflush (retrograde)".
- Meter for the material type/BOM item of the affected material:
  Configure meter like MDE meters.
  Option "compensation with material" = yes
  Option "material type" = material type of the material that is consumed
  or
  Option BOM item = BOM item of the material that is consumed

If the BOM item is used within meter configuration, it is important that within the OP's component list the material is always used as the same BOM item (from ERP work plan).

Posting/result

The automatically recorded quantity is deducted from the remaining quantity of the input batch and then the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.

22 Discrete Consumption Input

Usage

Key input values when collecting shop floor data are times and quantities. While times or durations are used to describe the time effort that was required to manufacture a material, quantities document the entire scope of the produced material. The objective is to process or output the material quantities defined in the order or the operation

In most cases, the quantities entered are sufficient to be able to execute the actions that will change stock quantities in the upper-level ERP system accordingly:

- Goods receipt from production
  The finished quantity will increase stock in the ERP system.

- Goods issue from production
  Based on the finished quantity, the consumption of the material that flowed in (so-called input material) can be calculated in reverse order of the production process in the ERP system, accounting for the bill of materials the order is based on. This will lead to a reduction of stock in the ERP system.

However, this kind of consumption calculation is oftentimes not enough to ensure a "clean" inventory management in the upper-level ERP system. Instead, there is a need to discretely enter order-related material consumption and to then post the consumption later in the ERP system.

You make use of this functions package if you would like to enter material consumption discretely and without a batch reference and upload it to an inventory management system.

Integration

The function can be integrated into a system dedicated to shop floor data collection (BDE), or also be used within the context of material and production logistics (MPL/ TRT).

These functions are used:

- Functions for discretely entering order-related material consumptions without a batch reference.
- Posting dialog at the Windows Terminal AIP to input discrete consumptions relating to material components during operation logoff, OP interruption or partial confirmation/upload.
- Entry based on the produced and manually entered total quantity or yield and/ or scrap.

- Providing material consumptions for confirmation/upload from HYDRA to the inventory management system in HYDRA standard format (requires that the interface used to upload material and batch data is licensed and activated).

23 Discrete Consumption Input at AIP

	→	

Summary

The function described below makes it possible to enter material consumption at the AIP shop floor terminal so that it can be uploaded via an interface in the form of goods movements.

Material consumption can be entered at the AIP shop floor terminal while an operation is being interrupted, while an operation is being logged off or when entering a partial confirmation/upload

Configuration

The document dealing with the configuration is to be taken into account for discrete consumption input.

Dialog and procedure

AIP provides the following features to enter material consumption based on the produced (total) quantity.

In the dialogs

- Log operation off (A_AB)
- Interrupt operation (A_UN)
- Partial confirmation/upload for operation (A_TR)

an additional "consumption" button is available with which the "Component consumption posting" dialog can be called up.

It is also possible to make the button available that is used to call up the "Component consumption posting" dialog from the MPL specific dialogs used for operation interruption (A_UN_MPL, A_UN_RF, A_UN_RS) or for operation log off (A_AB_MPL, A_AB_RF, A_AB_RS). However, what needs to be considered in this regard is that in MPL, consumption of batch-related material is posted differently.

Figure: "Partial confirmation/upload (A_TR)" dialog with "Consumption" button in AIP

The input dialog opens after the "Consumption" button is pressed.

The status information shown includes the workplace displayed in the dialog from which the function is called up, the operation and also the yield and scrap quantities entered in the dialog from which the function is called up. When calling up from an MPL dialog, the entered quantity is displayed based on whether classified ("Quality") as scrap or as yield.

Furthermore, a table is displayed showing the components that are flagged in the component list as consumption type "D". The following data is displayed in the table.

BOM item

BOM item

Input.Mat.No.

Material number of the material component.

Input.Mat.Des.

Material designation of the material component.

Consumption

Calculated consumption based on the quantity entered in the dialog from which the function is called up:

$$\text{Rechnerischer Verbrauch} = (\text{Gutmenge} + \text{Ausschuss}) * \text{Einsatzmenge}$$

Only yield and scrap quantities are taken into account when calculating the consumption.

Quantities are not set off against each other (e.g. scrap set off against yield) when the calculated consumption is determined.

Unit

Quantity unit (unit of the input quantity).

Input quantity

Input quantity required to produce one quantity unit of the output material.

When a component is selected from the list, the calculated consumption is proposed in the "Consumption" input field. If the actual consumption deviates from the calculated consumption, the operator can now modify this quantity; it is transferred to the list. Any consumption that was modified manually is shown highlighted in "green" in the list (e.g. 1,000 KG).

By pressing the "Reset" key, the component list can be called up again; this will recalculate the consumption quantities; any consumption quantities that were already modified will be overwritten.

After confirming this dialog and the dialog from which the function is called up, the consumption quantities are updated and are thus transferred to the server via the posting . There, the consumption is updated as the status in the component list at the operation. In addition, one material movement (goods issue) is written for each component.

The material consumption is only posted as a goods movement if the dialog was called up explicitly. Material consumptions that were not posted are not automatically posted when the operation is logged off or interrupted.

If the consumption input dialog is integrated in the partial upload dialog and if it is not called up when a partial upload is executed or when the entry is interrupted, the system will remember the calculated consumption for all material components of the consumption type "D" and will account for it the next time the consumption input dialog is called up.

Example (assumption: a material component with the input quantity 2 is defined at the operation):

- Call up the partial upload dialog (A_TR): Enter yield 9, scrap 1
- Call up the consumption input dialog (A_VERB)
    - (9 * 2 + 1 * 2 =) 20 is proposed as material consumption
    - Cancel the dialog (do not click on OK to confirm)
- Confirm the partial confirmation/upload dialog (A_TR)
    - The yield 9 and the scrap 1 are posted at the operation.
    - The system remembers the calculated consumption of 20
- Call up the partial upload dialog (A_TR) once more: Enter yield of 5
- Call up the consumption input dialog (A_VERB)
    - (20 + 5 * 2 =) 30 is now proposed as material consumption
    - Confirm with OK
- Confirm the partial confirmation/upload dialog (A_TR)
    - The yield 5 is posted to the operation
    - The consumption of 30 is posted as a goods movement

If after calling up the consumption input dialog it is canceled and thus closed and the quantity is then modified in the dialog that calls up the function and if then the consumption input dialog is once again called up, then the consumption quantities that were proposed the first time it was called up will still be shown in this dialog. This is why the proposed material consumption will need to be updated by clicking on the "Reset" button as the case may be.

24 Configuration of Discrete Consumption Input

	→	

Summary

Material consumption can be entered at the AIP shop floor terminal while an operation is being interrupted, while an operation is being logged off or when entering a partial confirmation/upload

System configuration

The relevant utilization material must be assigned to the operation in order to discretely record consumption. The meaning of the following fields must be observed in particular (logical consideration). The material components can be assigned in HYDRA manually (Order management: edit components) or from the upper level system

Material number

The material number uniquely identifies a material. In the component list, a material is only unique in association with a BOM item.

BOM item

The BOM item combined with the material number forms a unique key for a material component at an operation.

Material designation

The material designation is used to describe a material more exactly. It functions as commentary and is displayed in the input dialog.

Material category (type)

The material category controls, among other things, the way a material is technically processed in the HYDRA system. Relating to the material consumption input (not batch-related), the following value is important:

"M" Material component, should be entered in terms of consumption

Material type

The material type is another parameter used for processing control in HYDRA. The material type is used to control whether goods movement created from the material consumption should be uploaded to the ERP system or not. Unless defined otherwise assign the material type SYSTEM here.

To assure that a goods issue is uploaded via the interface, the option "Transfer to interface" must be set at the material type that the utilization material is based on.

When launching HYDRA for the first time or during discrete consumption input, it will need to be coordinated whether or for which material components consumption should be uploaded to the ERP system. If the MPL module is not used, this configuration is to be made during the HYDRA customizing process.

Consumption type

Set the consumption type to "D" (discrete consumption input) for material components that this function is used for to collect material consumption.

It is possible to transfer the consumption type using the relevant component segment of the ERP interface (EIS-ERP). When using the HYINFO interface as a part of the PP-PDC interface, the consumption type is made available by customizing the system accordingly.

The interface necessary to upload material consumption is not a part of this function.

Input quantity

The input quantity is a component quantity needed to manufacture one unit (one piece, for example) of the output material (article/item) being produced. It is used to calculate the theoretical material consumption based on the produced quantity.

Unit

Quantity unit of the material component in which material consumption is recorded.

Dialog configuration

In addition to making the necessary programs available, please also consider the following:

- Activate the dynamic dialog A_VERB, accounting for existing terminal groups in some cases
- Integrate the button used to call up the consumption input dialog in the relevant posting dialogs. The following configurations are possible here, whereas a) can be seen as an alternative to b) and c):
    a. Integration into the workflow step WF_AA_QUA.
    The button is found in the standard dialogs
        - Interrupt OP (A_UN),
        - Log off OP (A_AB), and
        - Partial confirmation/upload (A_TR)
        - in each workflow step in which quantities are entered manually.

    b. Integration into the workflow dialog WF_AUN_CHK
    The button is available in the standard dialog Interrupt OP (A_UN) in the confirmation workflow step.

    c. Integration into the workflow dialog WF_AAB_CHK
    The button is available in the standard dialog Log off OP (A_AB) in the confirmation workflow step.
- Integrating or checking the grid configuration for the dialog A_VERB (ctaiplay.ini)

- Optional: Uploading the material consumption as goods movement
  - Determining which material type is transferred to the material component from the ERP system (typically SYSTEM)
  - Setting the flag "Transfer to interface" at the material type. Options:
    - Directly via the master data configuration (only possible, if MPL/ TRT is active)
    - Per SQL to the HYDRA server (hysql -r -): update hz_typen set we_ext_kz = 'J' where hz_typ = 'SYSTEM';
    - Per SQL via the SQL tester in MOC: update hz_typen set we_ext_kz = 'J' where hz_typ = 'SYSTEM'
    - Per BAPI to the HYDRA server (Observe system!): hymwb -u9999 -c"DLG=MATTYP.UPDATE|MATTYP.MATTYP=SYSTEM|MATTYP.WEEXT:GEN=J|DAT=today|ZEI=now|"
  - Check per SQL: select hz_typ, we_ext_kz from hz_typen
  - Activate the material upload interface (see relevant documentation).